

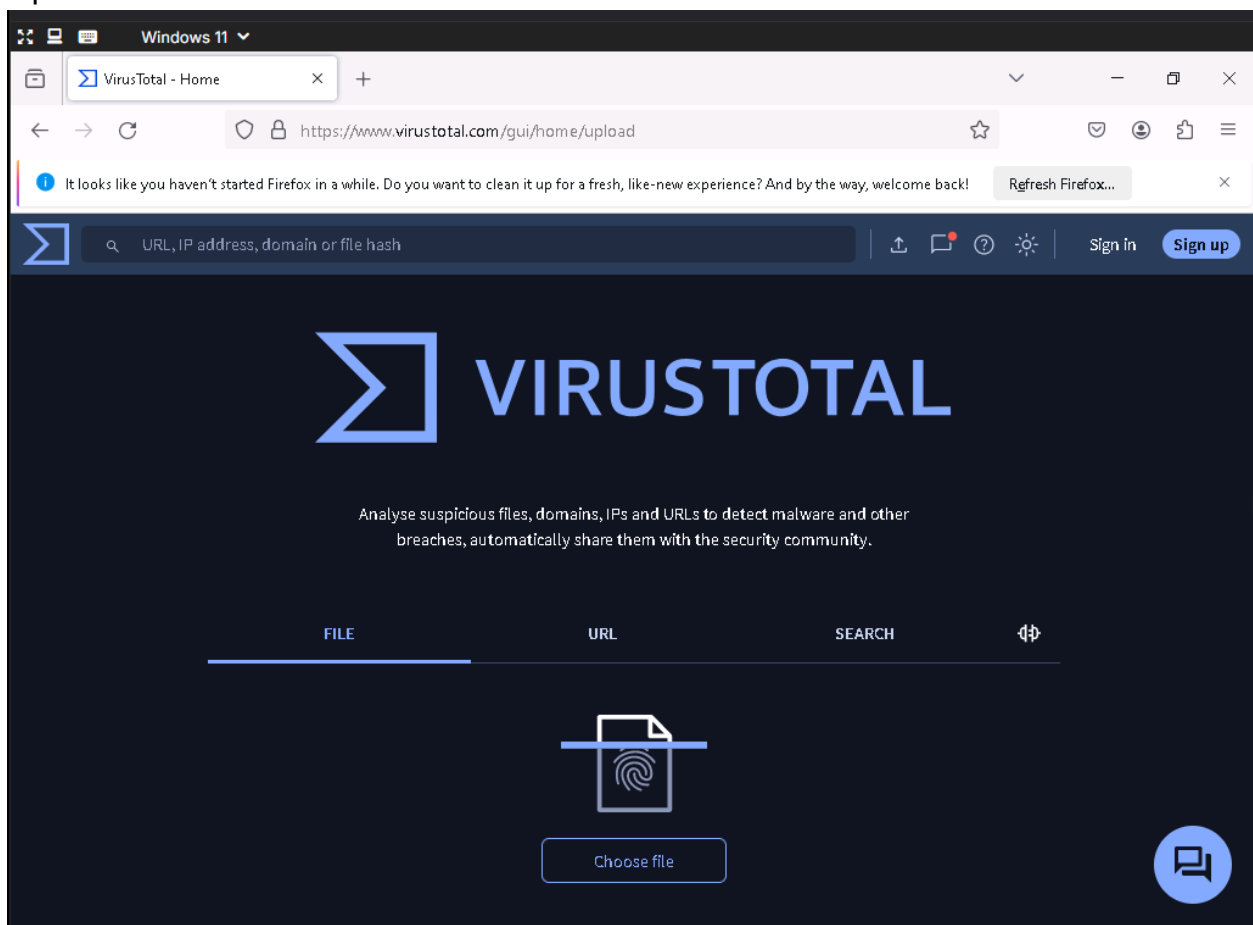
PROJECT: THREAT INTELLIGENCE ANALYSIS

Exercise 1: Data Collection through Malware Analysis.

TASK 1: Analyze Malware using VirusTotal and Hybrid Analysis.

VirusTotal is a free service that analyzes suspicious files and URLs. It generates a report that provides the total number of engines that marked the file as malicious, the malware name, and, if available, additional information about the malware.

Open VirusTotal on Windows:



Upload file wikiworm.exe

VirusTotal - File - 9fd30bda0edf3b10b326703303fa15995a688d200582822ef49422ebac87b7f7

https://www.virustotal.com/gui/file/9fd30bda0edf3b10b326703303fa15995a688d200582822ef49422ebac87b7f7

It looks like you haven't started Firefox in a while. Do you want to clean it up for a fresh, like-new experience? And by the way, welcome back!

Refresh Firefox...

9fd30bda0edf3b10b326703303fa15995a688d200582822ef49422ebac87b7f7

Sign inSign up

54 / 70

Community Score

54/70 security vendors flagged this file as malicious

ReanalyzeSimilarMore

9fd30bda0edf3b10b326703303fa15995a688d200582822ef49422ebac87b7f7

Size4.00 KB

Last Analysis Date8 days ago

EXE

peexe

detect-debug-environment

checks-user-input

long-sleeps

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY15

Join our Community

and enjoy additional community insights and crowdsourced detections, plus an API key to **automate checks**.

Code insights

This binary is a downloader worm. It downloads a random Wikipedia page to a local file named "downloaded.html". It then modifies this HTML file, injecting a link to a second-stage payload at http://people[.]freenet[.]de/artistsroom/information.exe and JavaScript code to automatically submit a form, likely for propagation. Finally, it uses CreateProcessA to launch Internet Explorer to open the modified local file, which triggers the download of the malicious executable. The string "Artwork by Second Part To Hell" is associated with the "Second Part To Hell" [Show more](#)

Details:

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY15

Join our Community

and enjoy additional community insights and crowdsourced detections, plus an API key to **automate checks**.

Basic properties

MD5	f68872773a88652541bd8d6ca9bda058
SHA-1	c8ac1b977d771c89ed7152f0daedb6a3e8b69b24
SHA-256	9fd30bda0edf3b10b326703303fa15995a688d200582822ef49422ebac87b7f7
Vhash	04303d165d1039zblz101fz
Authenticash	b21bb12e805bc702f34f54f93efebf05be5a88f2e423a90430ef5d41953d47f3
Imphash	ea8a76f6edad1ff842691f45567a5899
SSDEEP	48:ZvtQeJNRBDfUm8m0Zqaeo9bffs9lQ1eQC/g5eK09EHQfQFZduyvMstk:Z1HnxffuZDBWQ1kk09gPFZwyE
TLSH	T1B881768B8EA654E3E304A5751AE78943FF37D16018AECF1A0B25145D9D20A80F87DE8D
File type	Win32 EXE executable windows win32 pe peexe
Magic	PE32 executable (GUI) Intel 80386, for MS Windows
TrID	Win16 NE executable (generic) (32.2%) Win32 Executable (generic) (28.8%) OS/2 Executable (generic) (13%) Ge...

Community:

The screenshot shows a community interface for a malware image. At the top, there's a search bar with a long hash: `9fd30bda0edf3b10b326703303fa15995a688d200582822ef49422ebac87b7f7`. Below the search bar, two users are listed: **david.1** with the description "malware image" (dated 2022-09-22 05:54:56) and **mrpennypacker** with the description "livepage.apple.com" (dated 2022-06-02 04:28:54). A section titled "Voting details (9)" shows a grid of user votes. The grid contains nine entries, each with a user profile picture, name, time ago, and a vote (+1 or -1). At the bottom, there's a "Comments (4)" section and a blue icon for a document or report.

User	Vote
nasla (11 months ago)	+1
D00MsL4y3R (3 years ago)	-1
JaffaCakes118 (3 years ago)	-1
fhergom (5 years ago)	+1
levinho (5 years ago)	+1
gordong (6 years ago)	+1
tiaanalyst (8 years ago)	-1
rinnimathews (8 years ago)	-1
anonymous (13 years ago)	+1

Launching Hybrid Analysis on firefox

The screenshot shows the Hybrid Analysis website in a Firefox browser. The address bar displays `https://hybrid-analysis.com`. A notification bar at the top says: "It looks like you haven't started Firefox in a while. Do you want to clean it up for a fresh, like-new experience? And by the way, welcome back! Refresh Firefox...". Below the notification, there's a navigation bar with links: "Sandbox", "Quick Scans", "File Collections", "Resources", "Request Info", and "More". The main content area features the Hybrid Analysis logo (a red and blue circular icon) and the text "HYBRID ANALYSIS". Below the logo, there's a section with search options: "File/URL", "File Collection", "Report Search", "YARA Search", and "String Search". A large dashed box contains the text "Drag & Drop For Instant Analysis" and an icon of two interlocking gears.

HYBRID ANALYSIS

File/URL File Collection Report Search YARA Search String Search

Drag & Drop For Instant Analysis

Analysis of uploaded file:

 **HYBRID ANALYSIS**  [Request Info](#)

Analysis Overview

[Request Report Deletion](#) [Show Sample Content](#)

Submission name:	wikiworm.exe	malicious Threat Score: 100/100 AV Detection: 87% Labeled As: Trojan.Agent #4 #Comment #wikiworm #hi Post Link E-Mail 0 Community Score 0
Size:	4 KiB	
Type:	peexe executable	
Mime:	application/x-dosexec	
SHA256:	9fd30bda0edf3b10b326703303fa15995a688d200582822ef49422ebac87b7f7	
Submitted At:	2018-07-03 05:29:43 (UTC)	
Last Anti-Virus Scan:	2026-04-24 08:55:42 (UTC)	
Last Sandbox Report:	2023-12-06 22:34:31 (UTC)	

[Anti-Virus Results](#) [Updated 1 month ago - Click to Refresh](#)

CrowdStrike

MetaDefender

 **HYBRID ANALYSIS**  [Request Info](#)

Anti-Virus Results

[Updated 1 month ago - Click to Refresh](#)

CrowdStrike Falcon

Static Analysis and ML

Malicious (100%)

No Additional Data

MetaDefender

Multi Scan Analysis

Malicious (19/26)

More Details

Analysis Overview

Anti-Virus Scanner Results

Falcon Sandbox Reports (4)

Relations

Incident Response

Community (222)

Back to top

Details:

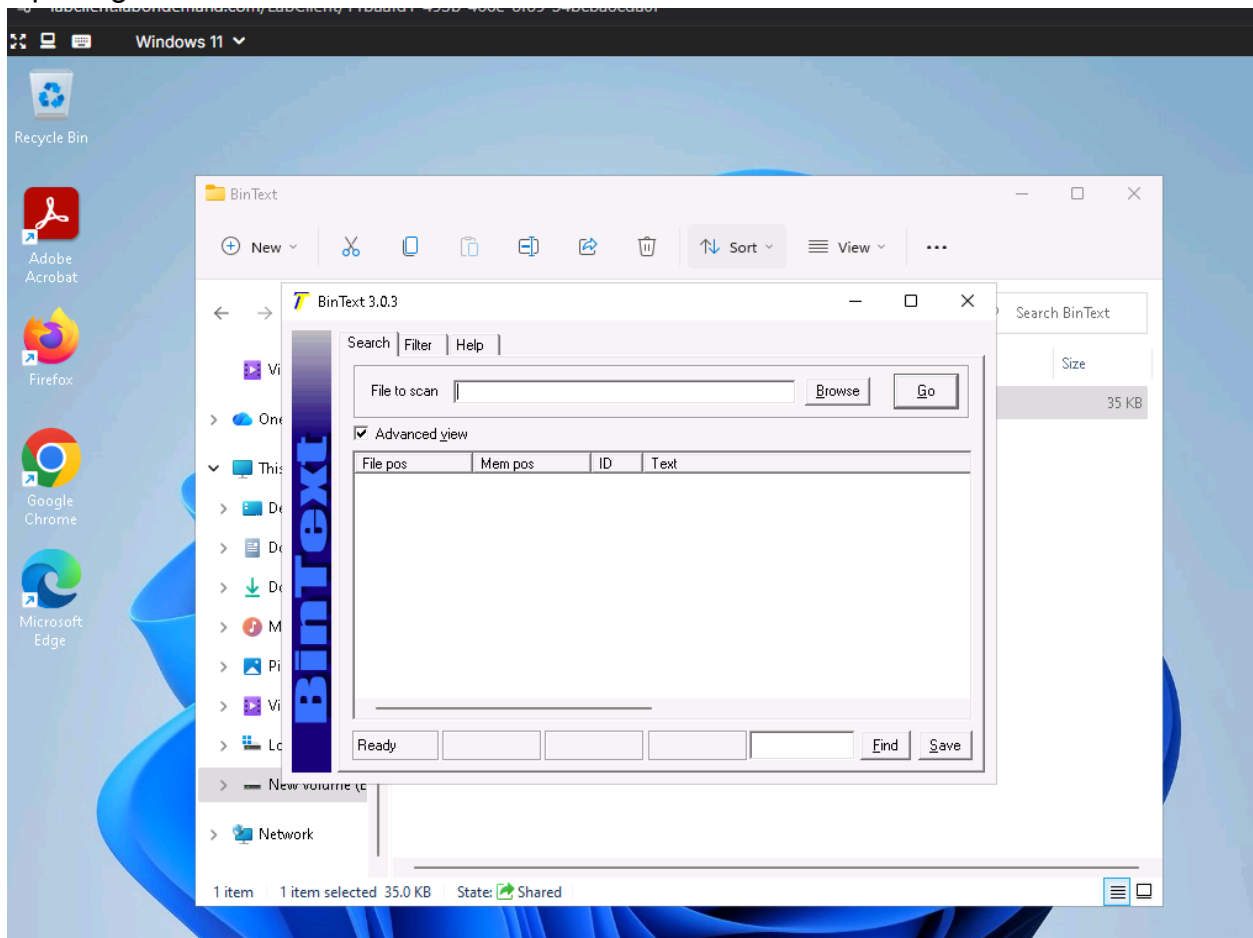
Anti-Virus Scan Results for OPSWAT Metadefender (19/26) ×			
Last update: 2026-04-24 08:55:42 (UTC)			
Vir.IT eXplorer	✗ Trojan.Win32.Agent.UN	K7	✗ Trojan (005f30271)
AhnLab	✗ Win-Trojan/Agent.4096.CA	Aurora	✗ Malware_-10
CMC	✗ Trojan_Win32_Malgent_MSR	Xcitium	✗ Malware
RocketCyber	✓	ClamAV	✗ Win.Trojan.Wicket-1
Huorong	✗ HVM:TrojanDownloader/Small.gen!A	Bitdefender	✓
Trellix	✗ GenDownloader.k	Gridinsoft	✓
Avira	✗ TR/Crypt.XPACK.Gen	Filseclab	✗ Trojan.Agent.un.ngz
Zillya!	✗ Trojan.AgentWin32.22894	Sophos	✗ Troj/Agent-CVP
VirusBlokAda	✓	NETGATE	✗ Worm.Win32.Wiki
TACHYON	✓	Varist	✗ W32/Downloader-Sml!Eldorado
			<button>Close</button>

This concludes the demonstration of malware analysis using VirusTotal and Hybrid
Analys

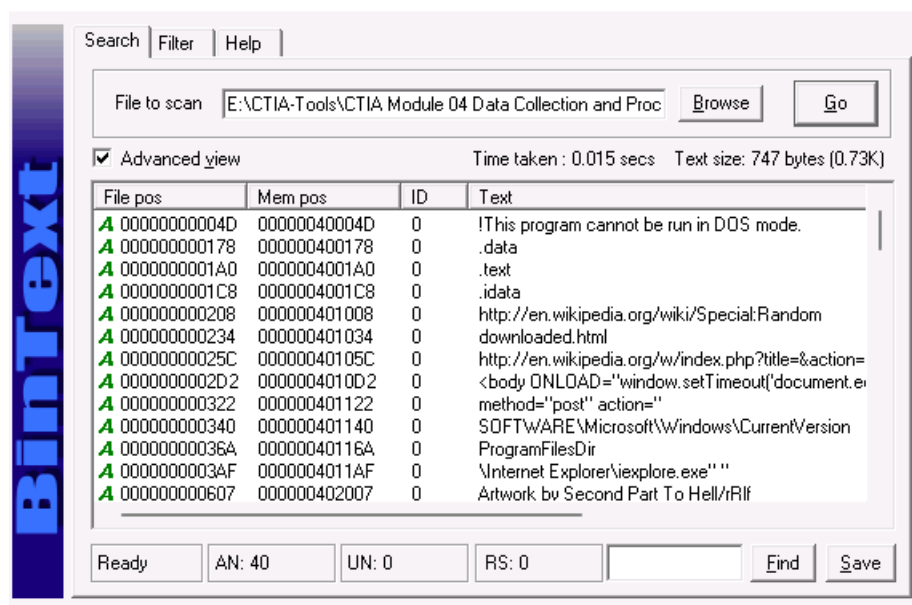
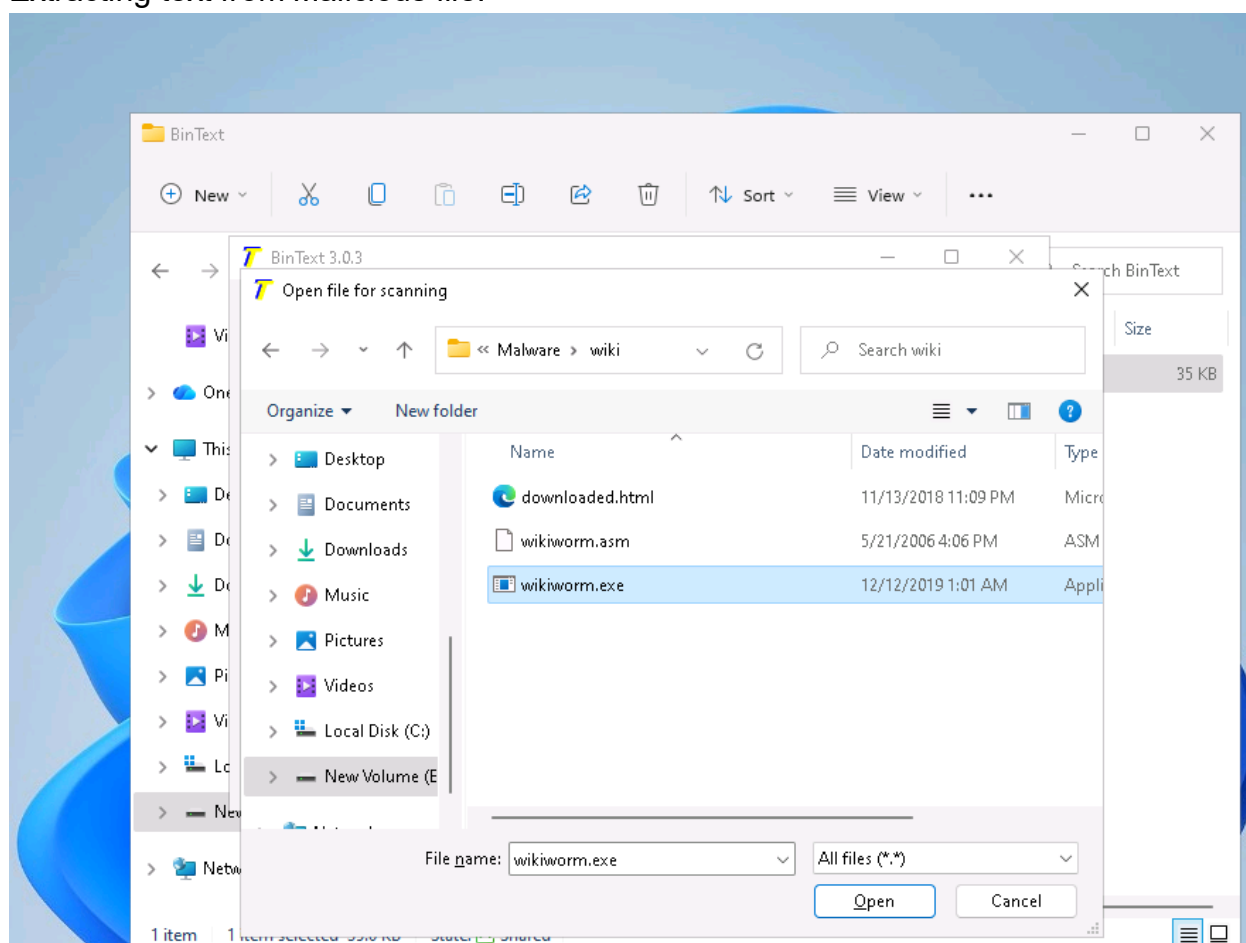
TASK 2: Analyze Malware using BinText

BinText is a text extractor that can extract text from any file. It includes the ability to find plain ASCII text, Unicode text, and Resource strings, providing useful information for each item.

Opening BinText on windows



Extracting text from Malicious file:



Exercise 2: Data Collection through Building Custom IoC's

The custom-built IoCs can be used by the analyst to collect internal as well as external data. Building custom IoCs not only assists in the development of threat intelligence but also helpful in disseminating the IoCs to the communities. These IoCs can also be shared between agencies and organizations for collaboration and enhancement by combining intelligence to achieve effectiveness in threat detection.

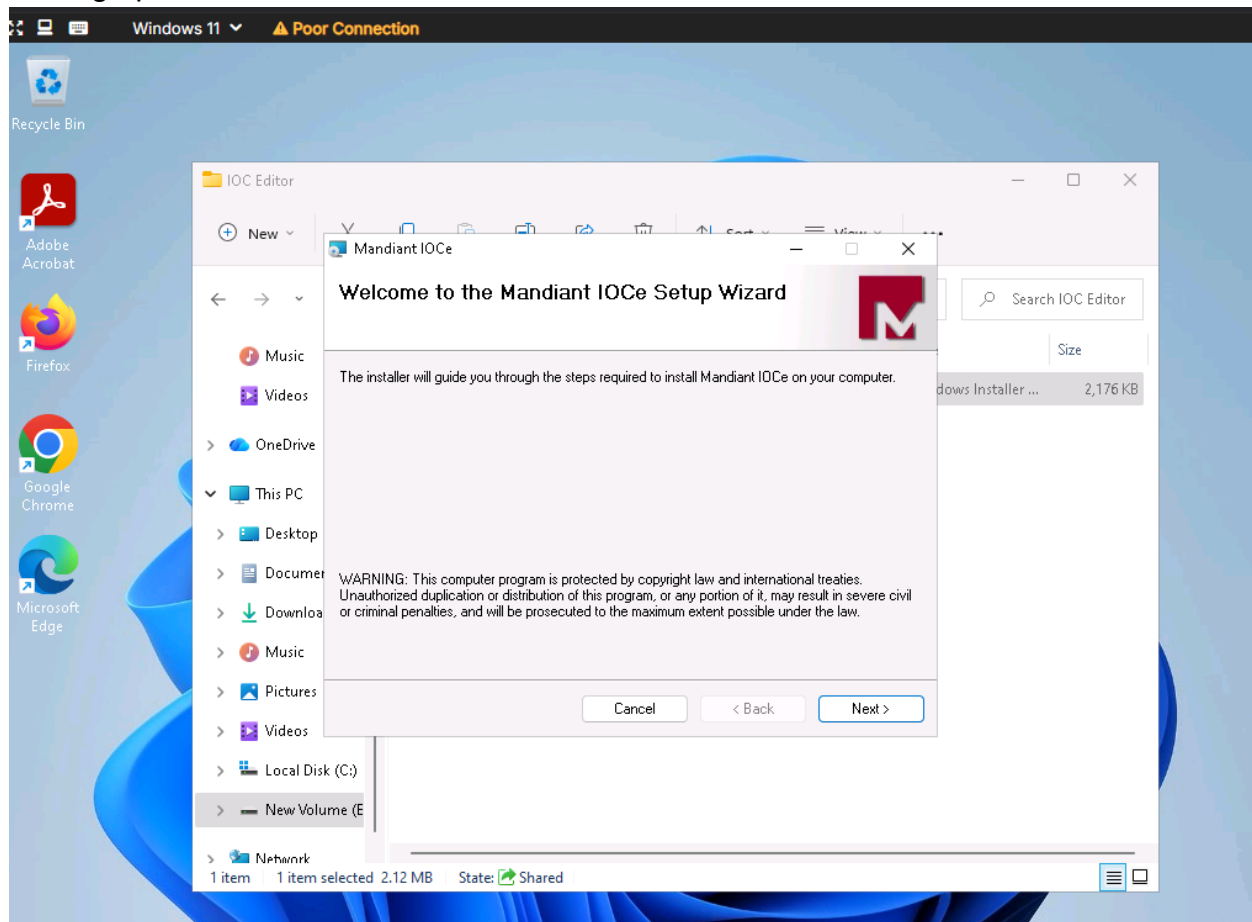
Lab Objectives

The objective of this lab is to help you understand how to build custom IoCs using IoC Editor.

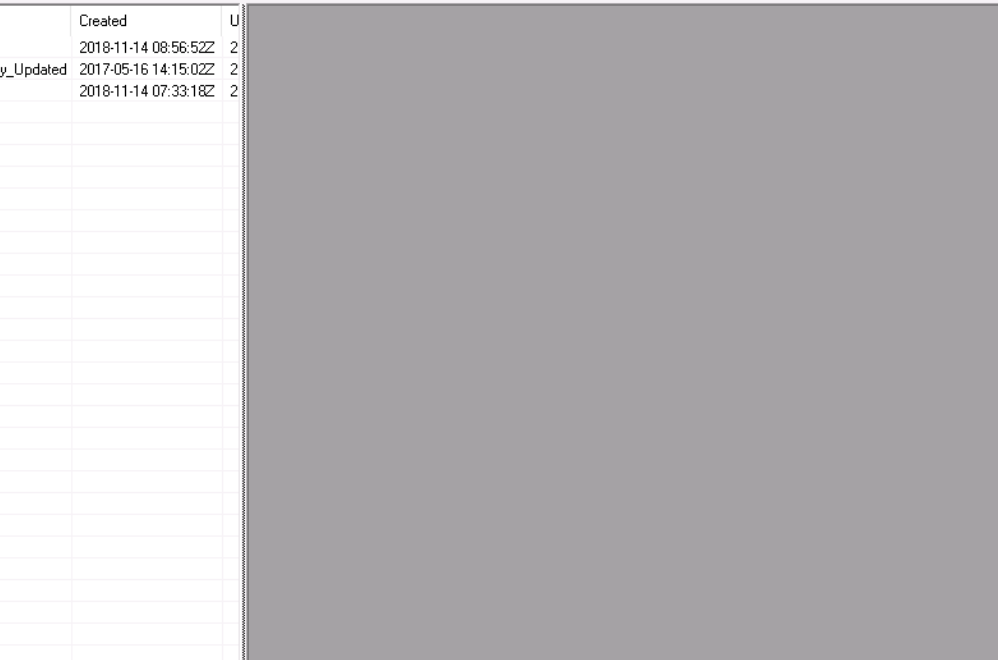
Overview of Building Custom IoC's

Analysts can build their own indicators based on particular patterns of observations and collect data from custom IoCs. These indicators can be built using the OpenIOC framework and then used with the extensible XML schema for scanning hosts for any signs of threats. They are used in an advanced scanning process to find specific types of vulnerabilities when they become active in the system.

Setting up Mnadiant loCe wizard on windows 11:



Launched Mandiant:



The screenshot displays the IOCe 2.2.0 application window. The title bar reads "IOCe 2.2.0 - E:\CTIA-Tools\CTIA Module 04 Data Collection and Processing\IOCs". The menu bar includes "File", "Search", "Tools", and "Help". The main content area features a table with three columns: "Name", "Created", and "U". The table contains three rows of data:

Name	Created	U
	2018-11-14 08:56:52Z	2
Ransom-WannaCry_Updated	2017-05-16 14:15:02Z	2
wikiworm	2018-11-14 07:33:18Z	2

The table is followed by several empty rows. The right side of the window is a large, empty gray area.

Reviewing file:

IOCe 2.2.0 - E:\CTIA-Tools\CTIA Module 04 Data Collection and Processing\IOCs

File Search Tools Help

Name	Created	U
	2018-11-14 08:56:52Z	2
Ransom-WannaCry_Updated	2017-05-16 14:15:02Z	2
wikiworm	2018-11-14 07:33:18Z	2

Name: Ransom-WannaCry_Updated

Author: shaag

GUID: 4838fc4d-aldf-4194-a364-f9c3578d3eca

Created: 2017-05-16 14:15:02Z

Modified: 2017-05-18 19:29:57Z

Description:

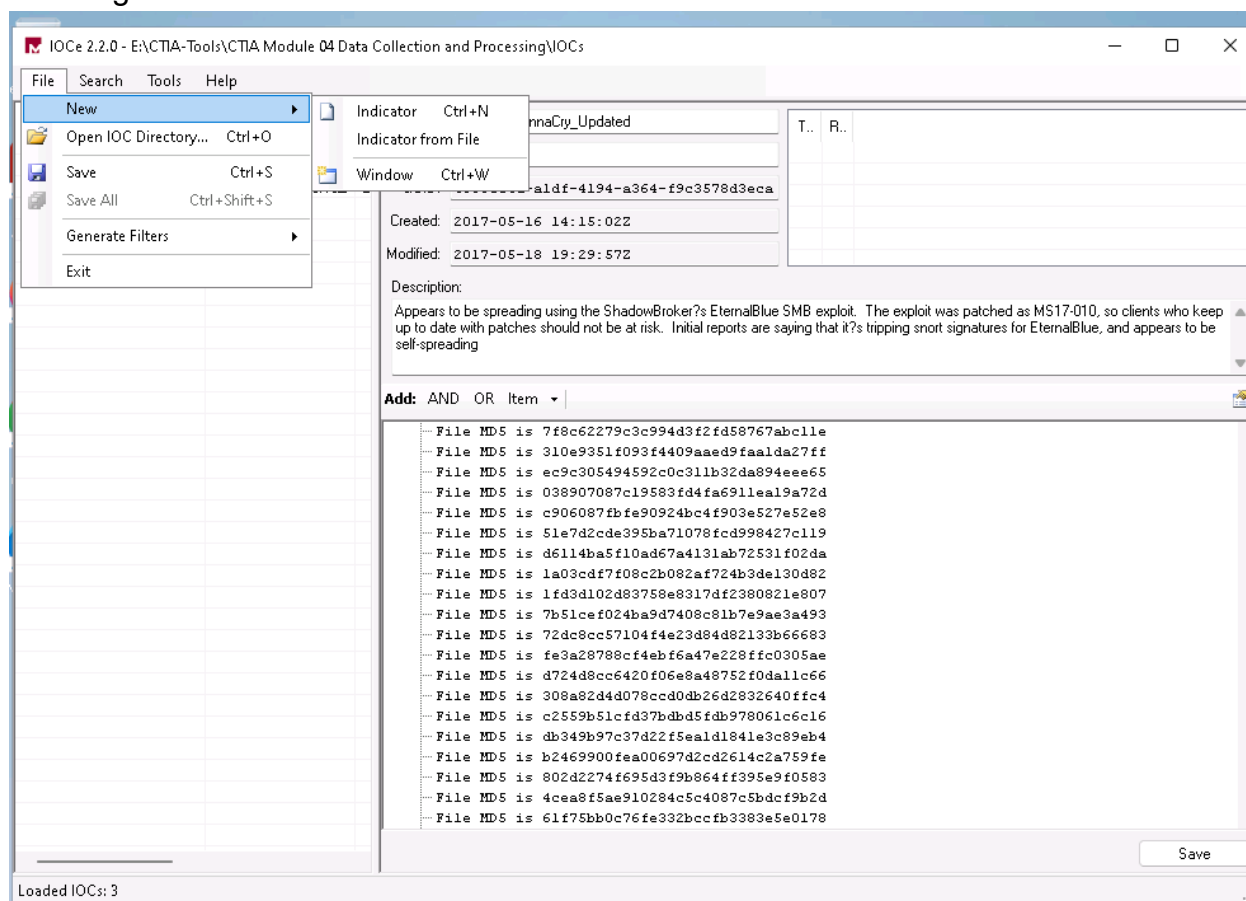
Appears to be spreading using the ShadowBroker's EternalBlue SMB exploit. The exploit was patched as MS17-010, so clients who keep up to date with patches should not be at risk. Initial reports are saying that it's tripping snort signatures for EternalBlue, and appears to be self-spreading

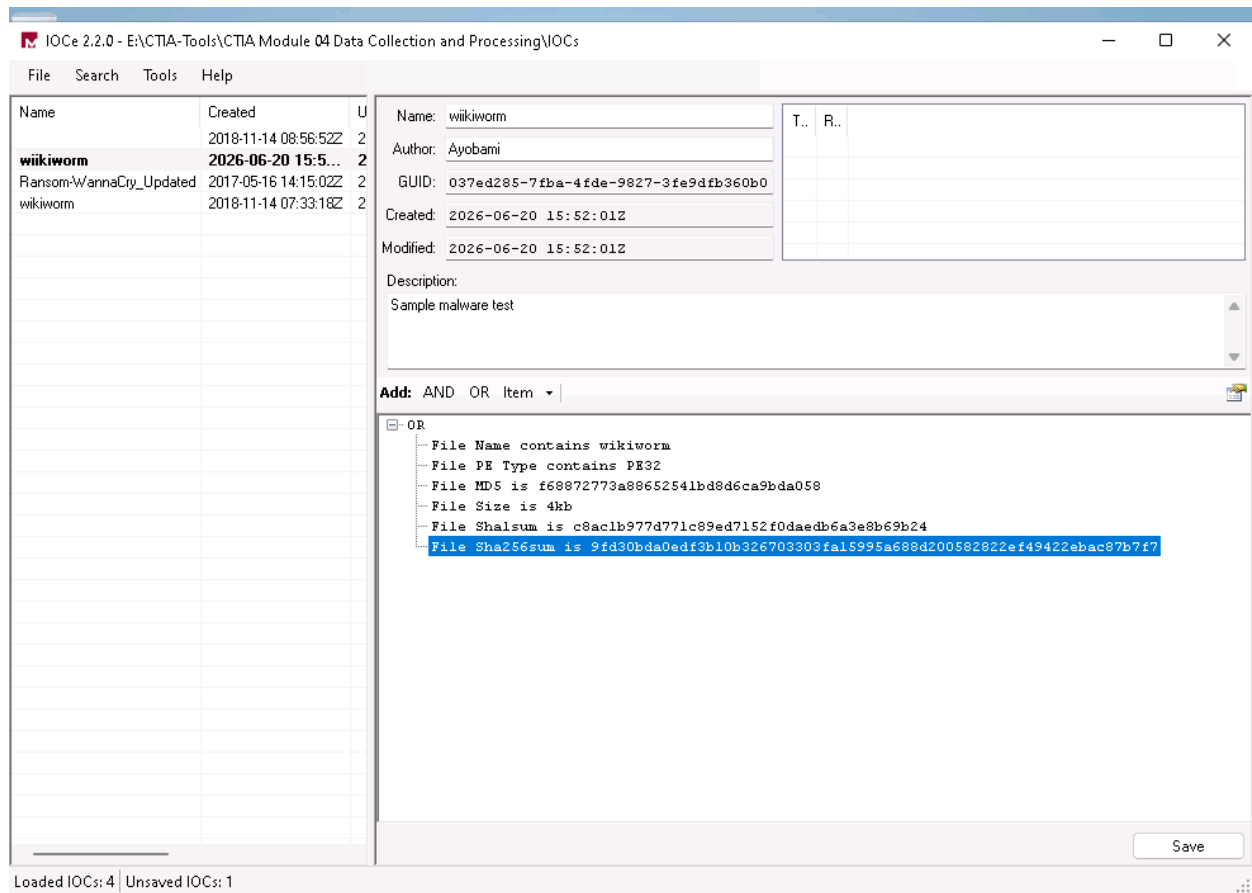
Add: AND OR Item

OR

- File Sha256sum is b9c5d4339809e0ad9a00d4d3dd26fdf44a32819a54abf846bb9b560d81391c25
- File Sha256sum is 75457f282337bccb7f48db927bdfb16d517c13eb5f361419776363bcd2a64f2
- File MD5 is 84c82835a5d21bbcf75a61706d8ab549
- File MD5 is 52e5764787dc2b4d490be75fea7f1f0d
- File MD5 is 998ea85d3e72824a8480d606d33540a6
- File MD5 is 7f7ccaal6fb15eb1c7399d422f8363e8
- File MD5 is a765434505308937e089d885c4f16d49
- File MD5 is 509c41ec97bb81b0567b059aa2f50fe8
- File MD5 is 340a0e61c7f9b4e17e66e5114b1fffd
- File MD5 is 765c0640f3d4942611b93692751adebc
- File MD5 is 25944f5853340835d4d9486c97205f58
- File MD5 is 42f6bfd62e0dbc0c7d50e03f4978122a
- File MD5 is 22b75c56c6a4851f27faed66ca8f000b
- File MD5 is 22e58018f9a0fe4a18250f33bclc53e0
- File MD5 is 28881174ae54d7ffd5f9cc0cbdc7b589
- File MD5 is 0de801a31186555329eafa05f6946e10
- File MD5 is 7dcb31fa5876b9b6ac0b5e8fa2980b3e
- File MD5 is 7de2e4f0db359886a8875faa982ad515
- File MD5 is 289f0815e35488e4e71688156148d83

Creating a new indicator:





In this task, I have successfully completed building custom IoC for wikiworm using IOC Editor.

Exercise 3: Data Collection through Python Scripting : Database Scrapping.

Python scripts are widely used programs for assessing threat data and adding more contextual information. Python's versatility, extensive libraries, and ease of use make it a prime choice for automating the process of gathering valuable information from various sources. Security professionals employ Python scripts to interact with diverse data repositories, such as logs, APIs, websites, and databases.

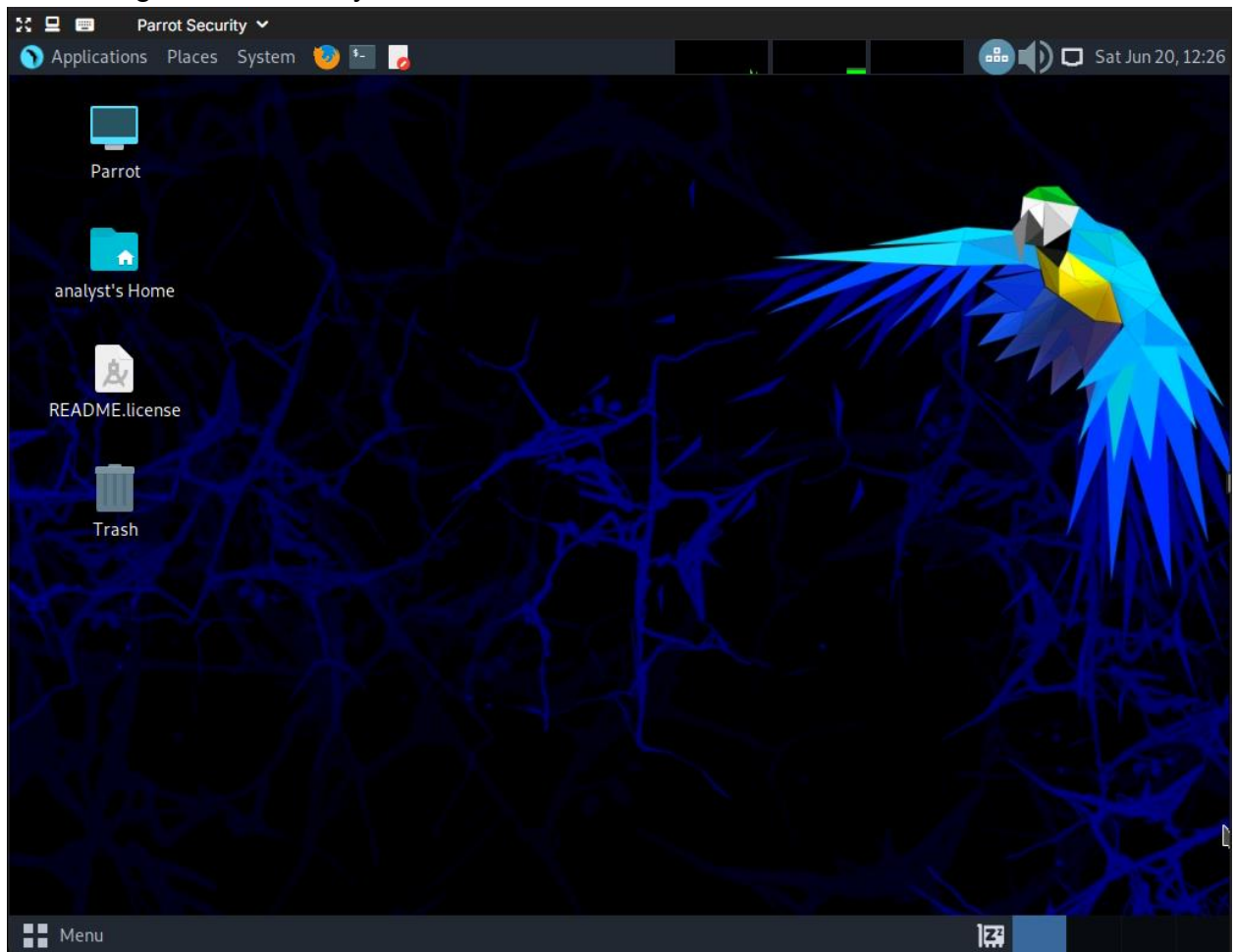
Lab Objectives

The objective of this lab is to help you understand how to collect data using Python scripting.

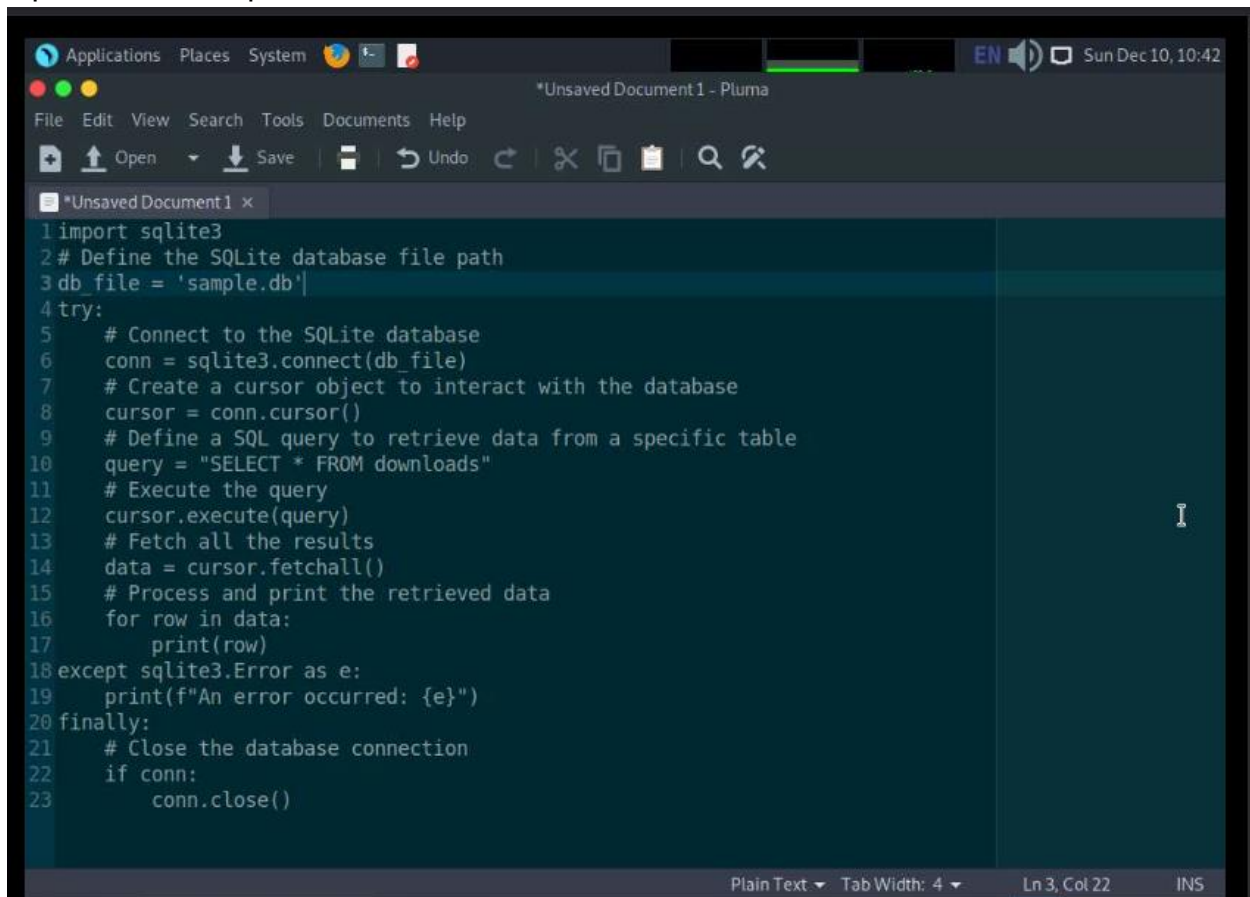
Overview of Python Scripting

Python scripts extract, parse, and structure the threat data, allowing an effective analysis and monitoring. Automation and scheduling capabilities enable regular updates, ensuring the threat data remains current. Python also plays a crucial role in data enrichment, helping to contextualize and enhance the collected information. With the ability to generate visualizations, alerts, and reports, Python scripting empowers cybersecurity experts to better understand the evolving threat landscape, respond swiftly to emerging risks, and bolster the security posture of organizations. Analysts need to specify the threat source or API to extract threat data, which may include a log file, an API endpoint, or a database

Launching Parrot Security VM:



Open Puma to input code:

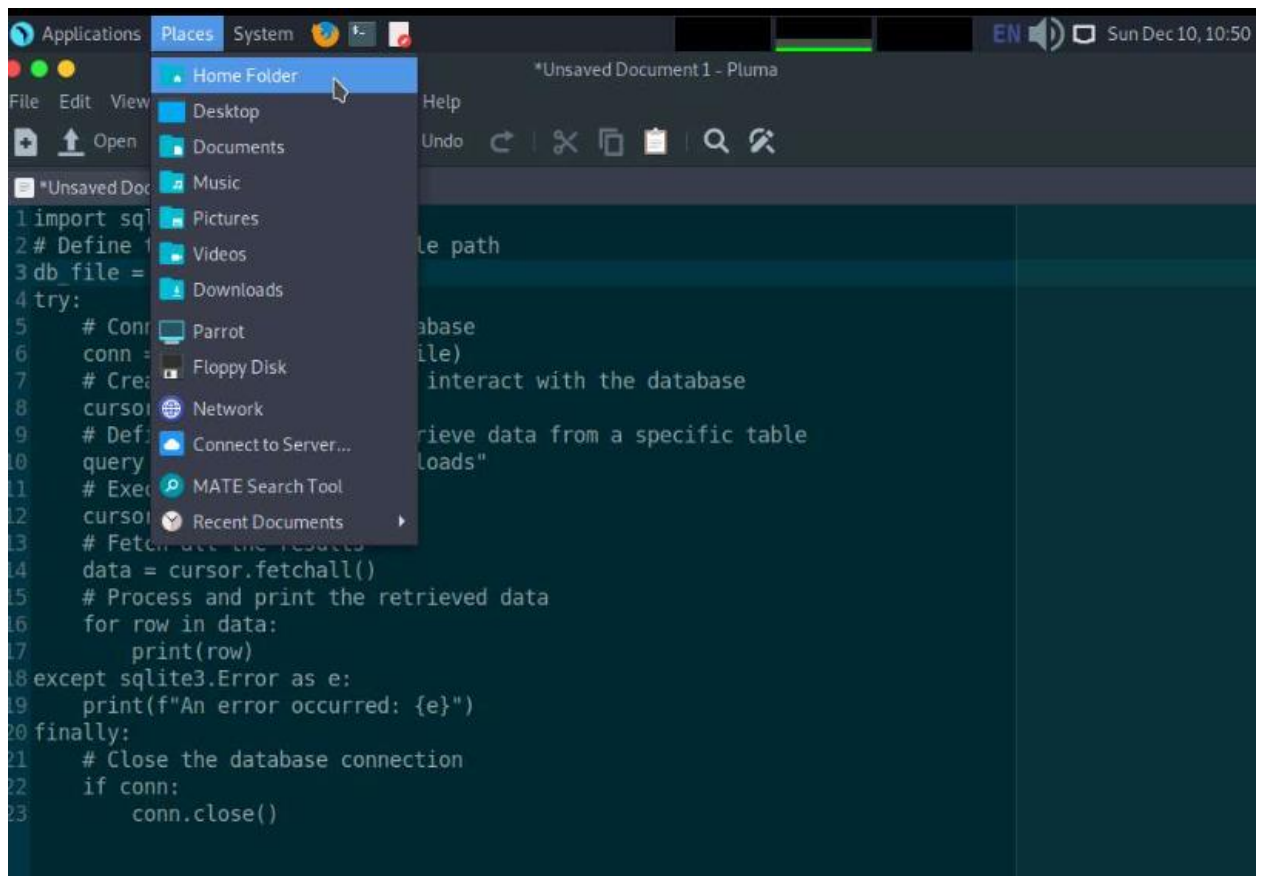


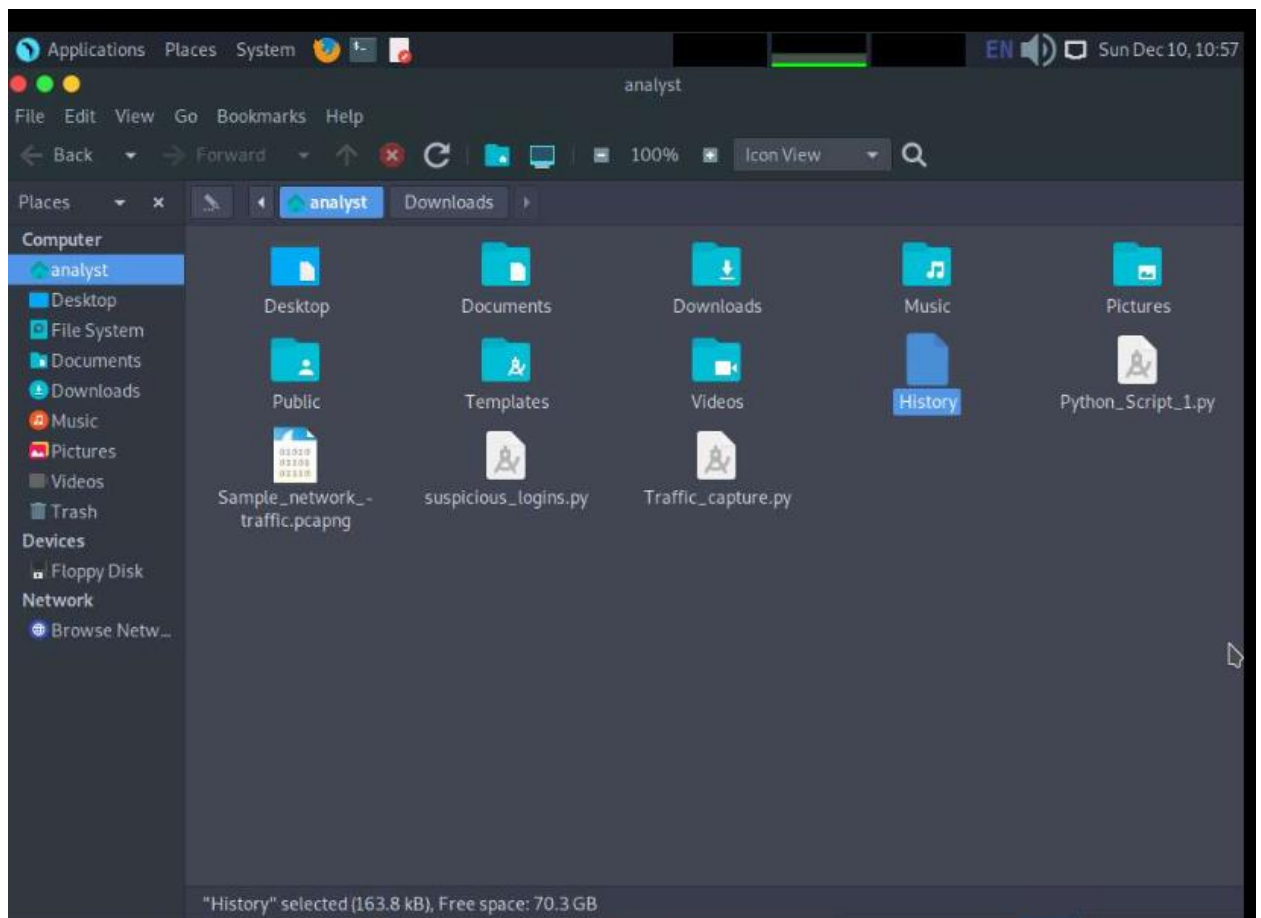
The screenshot shows the Puma text editor interface. The title bar indicates the file is '*Unsaved Document1 - Puma'. The menu bar includes File, Edit, View, Search, Tools, Documents, and Help. The toolbar contains icons for Open, Save, Undo, and other standard text editor functions. The main editing area contains a Python script that connects to an SQLite database, executes a query to retrieve data from a table named 'downloads', and prints the results. The script is as follows:

```
1 import sqlite3
2 # Define the SQLite database file path
3 db_file = 'sample.db'
4 try:
5     # Connect to the SQLite database
6     conn = sqlite3.connect(db_file)
7     # Create a cursor object to interact with the database
8     cursor = conn.cursor()
9     # Define a SQL query to retrieve data from a specific table
10    query = "SELECT * FROM downloads"
11    # Execute the query
12    cursor.execute(query)
13    # Fetch all the results
14    data = cursor.fetchall()
15    # Process and print the retrieved data
16    for row in data:
17        print(row)
18 except sqlite3.Error as e:
19     print(f"An error occurred: {e}")
20 finally:
21     # Close the database connection
22     if conn:
23         conn.close()
```

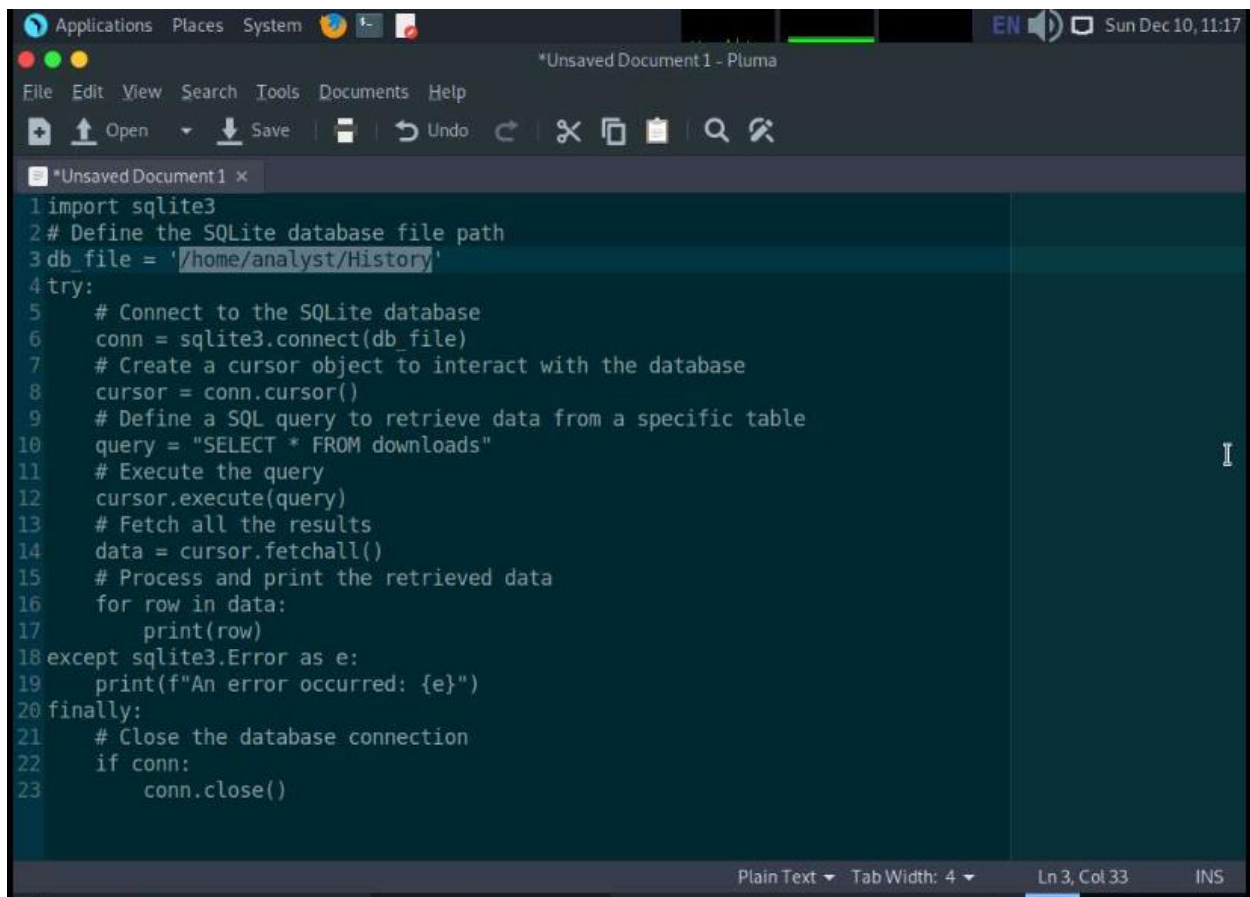
The status bar at the bottom shows 'Plain Text', 'Tab Width: 4', 'Ln 3, Col 22', and 'INS'.

1. we will analyze Google Chrome **History** SQLite database file for SQLite database scraping which is placed at **/home/analyst** location.





Enter File Location:

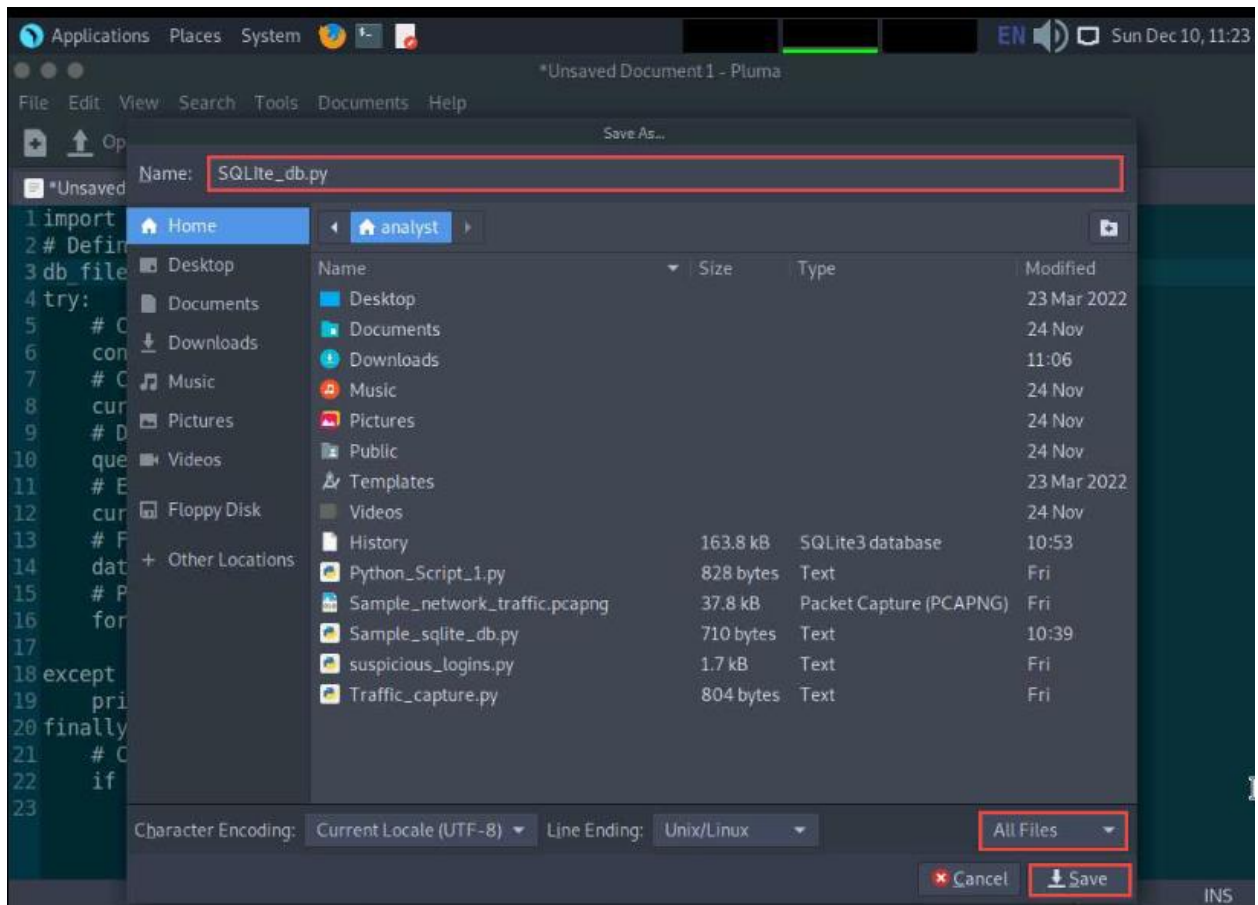


The screenshot shows a Pluma text editor window titled "*Unsaved Document 1 - Pluma". The window has a menu bar with "File", "Edit", "View", "Search", "Tools", "Documents", and "Help". Below the menu bar is a toolbar with icons for "Open", "Save", "Undo", "Redo", "Cut", "Copy", "Paste", "Find", and "Replace". The main text area contains a Python script that connects to a SQLite database, executes a query, and prints the results. The script is as follows:

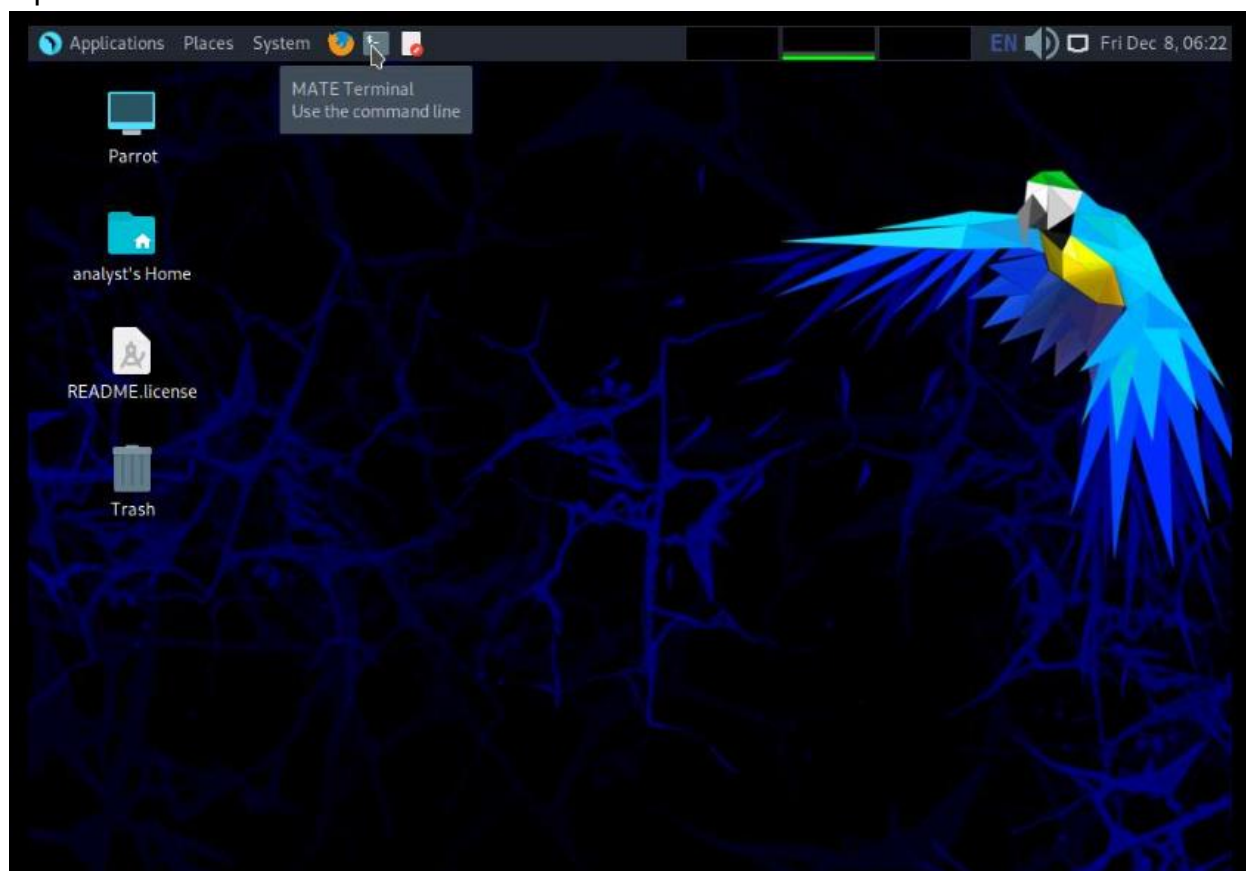
```
1 import sqlite3
2 # Define the SQLite database file path
3 db_file = '/home/analyst/History'
4 try:
5     # Connect to the SQLite database
6     conn = sqlite3.connect(db_file)
7     # Create a cursor object to interact with the database
8     cursor = conn.cursor()
9     # Define a SQL query to retrieve data from a specific table
10    query = "SELECT * FROM downloads"
11    # Execute the query
12    cursor.execute(query)
13    # Fetch all the results
14    data = cursor.fetchall()
15    # Process and print the retrieved data
16    for row in data:
17        print(row)
18 except sqlite3.Error as e:
19     print(f"An error occurred: {e}")
20 finally:
21     # Close the database connection
22     if conn:
23         conn.close()
```

The status bar at the bottom of the window shows "Plain Text", "Tab Width: 4", "Ln 3, Col 33", and "INS".

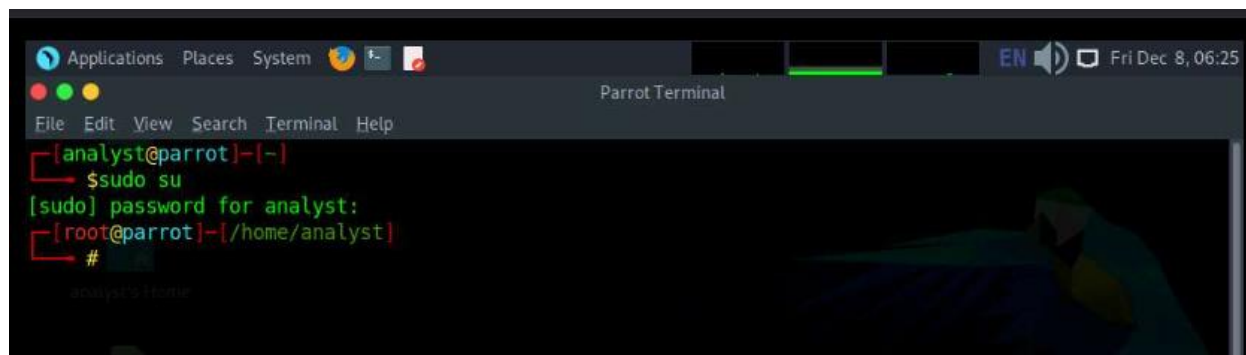
Save:



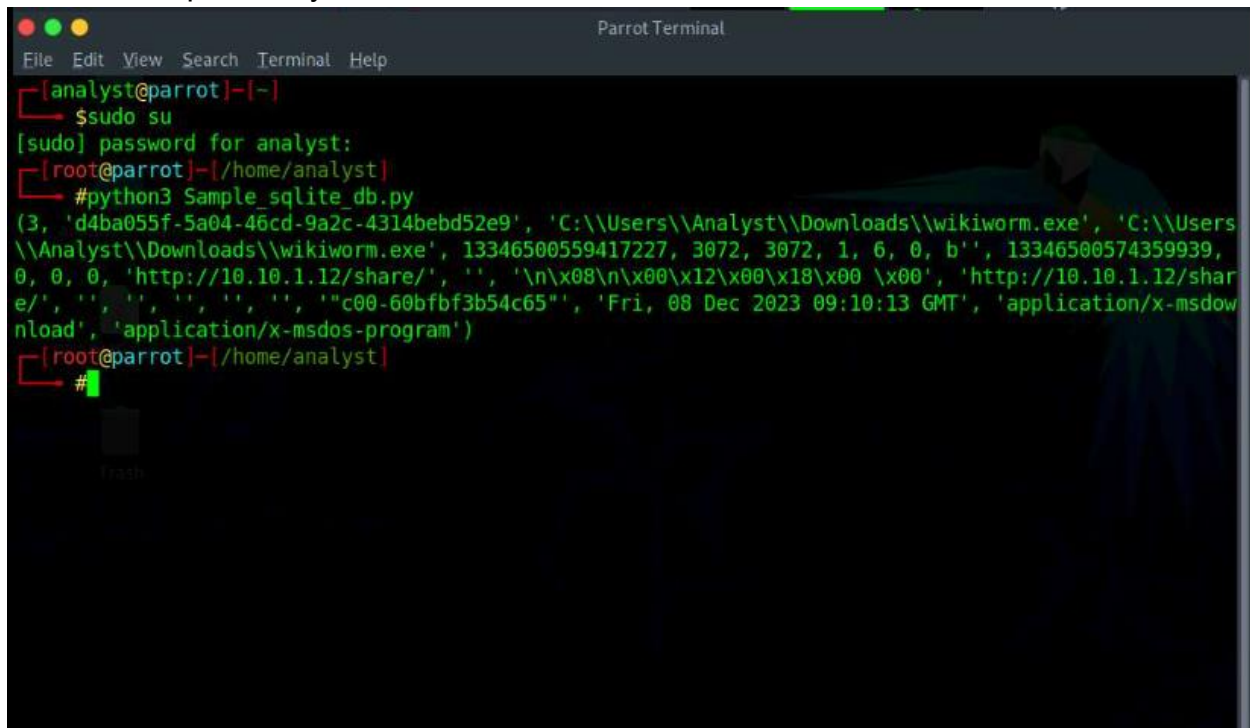
Open Mate Terminal:



Use Sudo:



Execute Script with Python:



```
Parrot Terminal
File Edit View Search Terminal Help
[analyst@parrot]~$ sudo su
[sudo] password for analyst:
[analyst@parrot]~$ #python3 Sample_sqlite_db.py
(3, 'd4ba055f-5a04-46cd-9a2c-4314bebd52e9', 'C:\\Users\\Analyst\\Downloads\\wikiworm.exe', 'C:\\Users\\Analyst\\Downloads\\wikiworm.exe', 13346500559417227, 3072, 3072, 1, 6, 0, b'', 13346500574359939, 0, 0, 0, 'http://10.10.1.12/share/', '', '\n\x08\n\x00\x12\x00\x18\x00 \x00', 'http://10.10.1.12/share/', '', '', '', '', '', '"c00-60bfbf3b54c65"', 'Fri, 08 Dec 2023 09:10:13 GMT', 'application/x-msdownload', 'application/x-msdos-program')
[analyst@parrot]~$ #
```

From the above screenshot we can see that the **wikiworm.exe** file was downloaded from **http://10.10.1.12/share** website using **Google Chrome** browser.

Exercise 4: Structuring/Normalization of Collected Data

Data collection is performed using various means and from various sources. However, the collected data may be in different formats or in the form of unstructured bulk data. Hence, to make the collected data usable and rich in intelligence, it needs to be in a structured form.

As an analyst, you must employ filtering, tagging, and queuing techniques to sort out the relevant and structured data from large amounts of unstructured data.

Structuring/normalization of data is performed to reduce the data redundancy, thus improving data acceptability and data integrity

Lab Objectives

The objective of this lab is to help you understand how to structure and format data using STIX. **STIX (Structured Threat Information eXpression)** is a standardized language used to **represent, organize, and exchange cyber threat intelligence (CTI)** in a structured, machine-readable format.

Overview of Structuring/Normalization of Collected Data

Normalization is the process of structuring unstructured data to make it sorted and usable by humans, as well as automated machine tools for intelligence consumption.

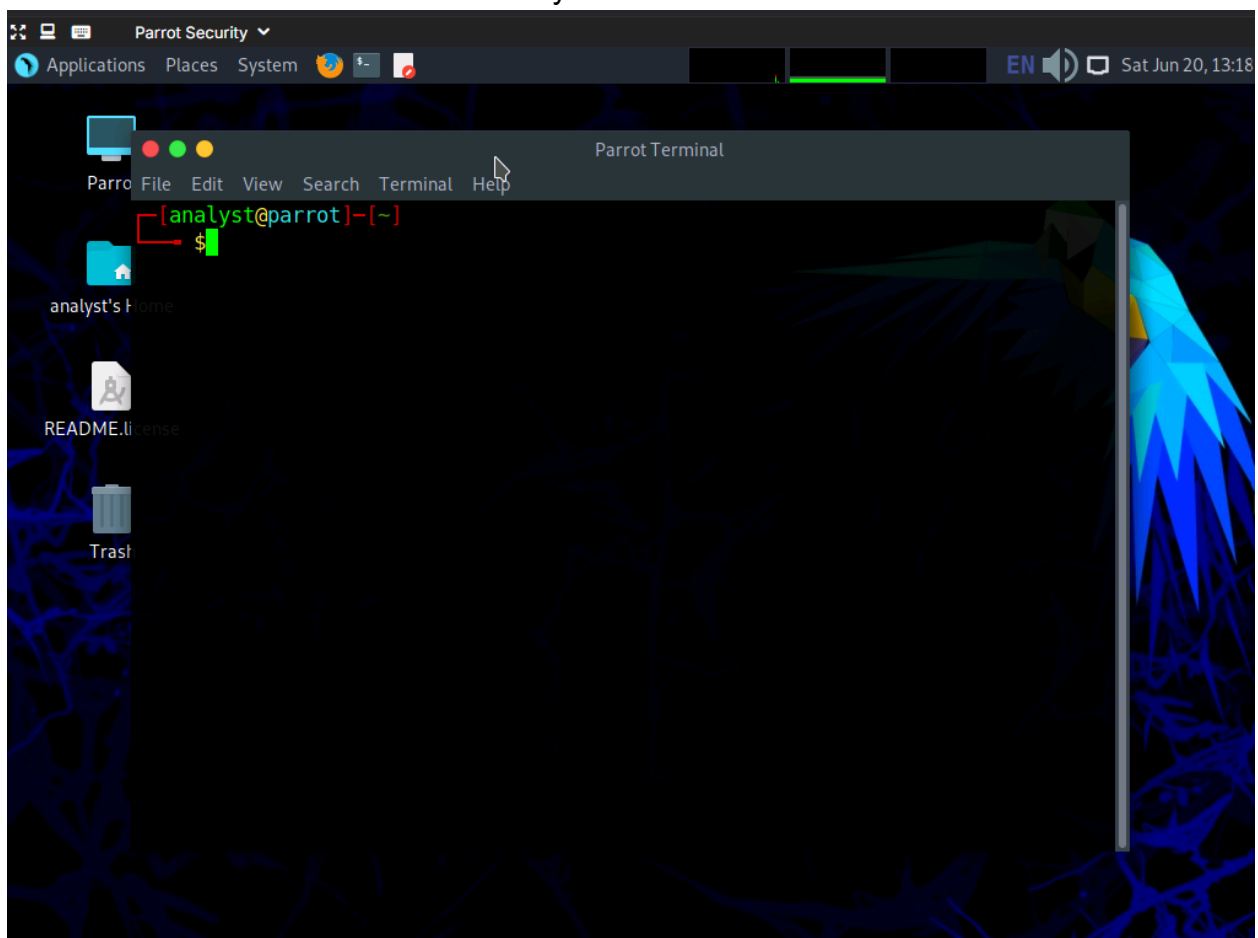
Structuring this collected bulk data is performed based on the following parameters:

- **Accuracy:** Depending on the accuracy of the collected data, the analyst can normalize the acquired data.
- **Reliability:** Reliability of the sources is key to good threat intelligence creation. Hence, data from more reliable sources is considered during the normalization process.
- **Usability:** Data that is relevant and contains precise information about the threats and vulnerabilities can be used with full efficiency.
- **Feature selection and extraction:** Specific organization require feature selection algorithms can also be employed to perform the normalization of the collected chunks of data.

- **Alert correlation:** By correlating collected data with well-known malicious samples or other specific alerts generated by the security system, analysts can perform the normalization of the data..
- **Anomaly detection:** Analysts can also normalize the collected data by examining it with logged security event information to identify known attack patterns or signatures and to identify anomalies for which an alert is generated showing the possible attack.

Tasks:

Lunch Mate Terminal on Parrot Security VM



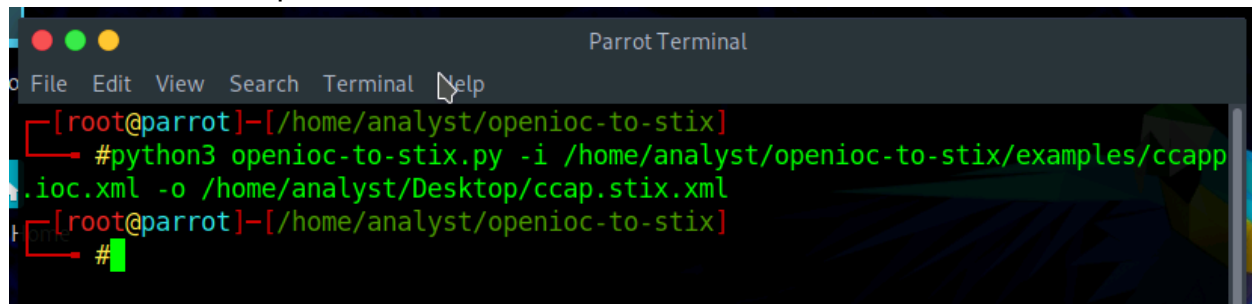
Navigating to openioc-to-stix directory

```
Parrot Terminal
File Edit View Search Terminal Help
[analyst@parrot]-[~]
$ sudo su
[sudo] password for analyst:
[root@parrot]-[/home/analyst]
# cd openioc-to-stix/
[root@parrot]-[/home/analyst/openioc-to-stix]
#
```

Installing stix

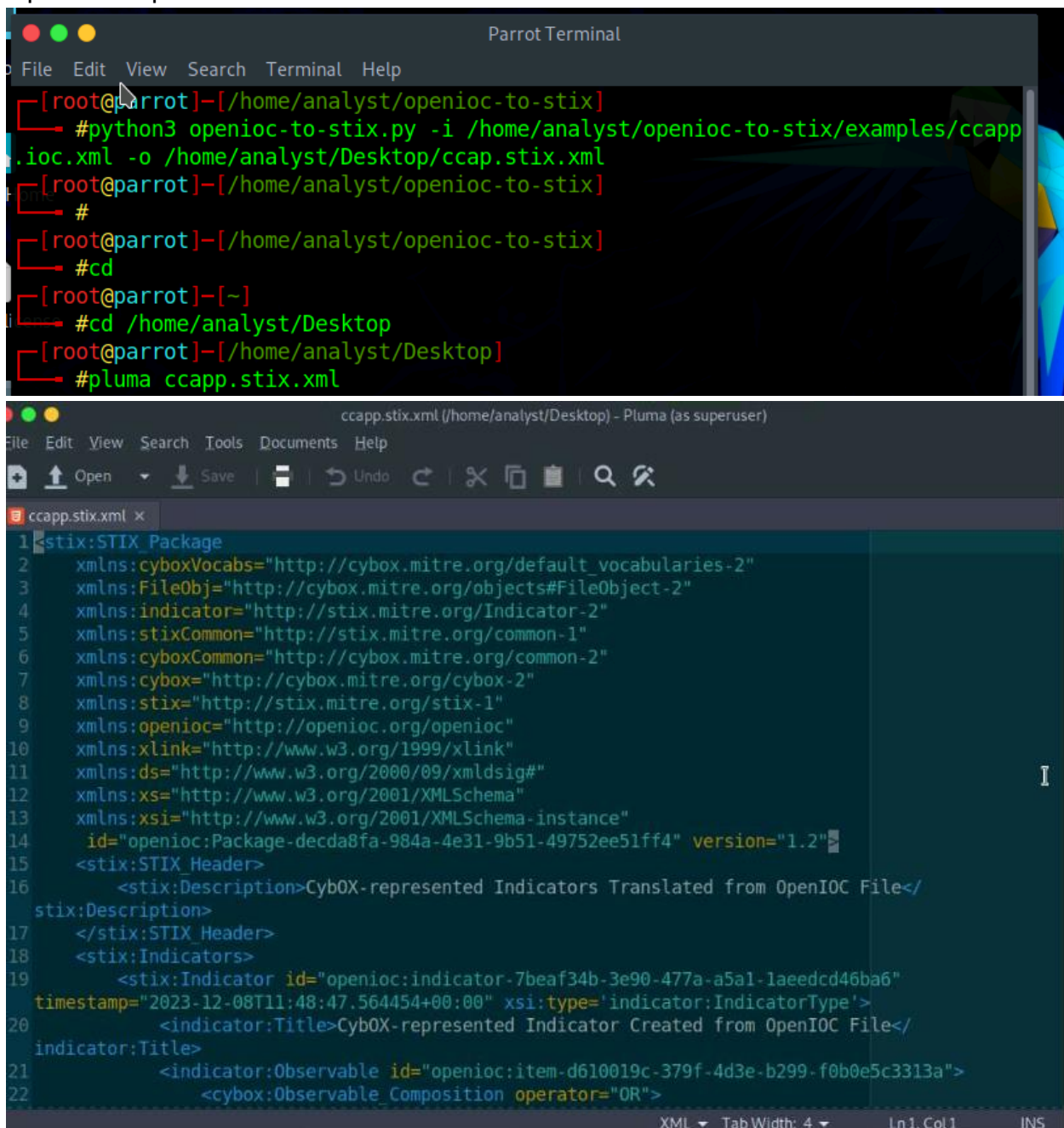
```
Parrot Terminal
File Edit View Search Terminal Help
Requirement already satisfied: python-dateutil in /usr/lib/python3/dist-packages
(from stix) (2.8.1)
Collecting mixbox>=1.0.4
  Downloading mixbox-1.0.5-py2.py3-none-any.whl (47 kB)
    | 47 kB 3.8 MB/s
Requirement already satisfied: lxml>=2.2.3 in /usr/lib/python3/dist-packages (fr
om stix) (4.6.3)
Collecting weakrefmethod>=1.0.3
  Downloading weakrefmethod-1.0.3.tar.gz (7.5 kB)
Collecting ordered-set
  Downloading ordered_set-4.1.0-py3-none-any.whl (7.6 kB)
Building wheels for collected packages: weakrefmethod
  Building wheel for weakrefmethod (setup.py) ... done
  Created wheel for weakrefmethod: filename=weakrefmethod-1.0.3-py3-none-any.whl
size=6989 sha256=1842e823a61e115fb4371d42c9635f362cffb96a292e1175ef85292a3757ed
91
  Stored in directory: /root/.cache/pip/wheels/45/89/7e/f8319c655f7ab62b5e52f5ff
62b3a10deac8ed23989ec53f51
Successfully built weakrefmethod
Installing collected packages: weakrefmethod, ordered-set, mixbox, cybox, stix
Successfully installed cybox-2.1.0.21 mixbox-1.0.5 ordered-set-4.1.0 stix-1.2.0.
11 weakrefmethod-1.0.3
[root@parrot]-[/home/analyst/openioc-to-stix]
#
```

Convert file from OpenIOC format to STIX format

A screenshot of a Parrot Terminal window. The title bar at the top says "Parrot Terminal" and has three colored window control buttons (red, green, yellow) on the left. Below the title bar is a menu bar with the options "File", "Edit", "View", "Search", "Terminal", and "Help". The terminal window has a dark background with a faint, stylized parrot logo on the right side. The prompt is "[root@parrot]-[/home/analyst/openioc-to-stix]". The command being entered is "#python3 openioc-to-stix.py -i /home/analyst/openioc-to-stix/examples/ccapp.ioc.xml -o /home/analyst/Desktop/ccap.stix.xml". The cursor is at the end of the command, and a green cursor block is visible on the line below.

```
[root@parrot]-[/home/analyst/openioc-to-stix]
#python3 openioc-to-stix.py -i /home/analyst/openioc-to-stix/examples/ccapp
.ioc.xml -o /home/analyst/Desktop/ccap.stix.xml
[root@parrot]-[/home/analyst/openioc-to-stix]
#
```

Open file in pluma editor



The image shows a Parrot Terminal window with a dark background and a Pluma text editor window. The terminal displays a series of commands being executed in a root shell on a parrot machine. The commands convert an OpenIOC file to STIX format using a Python script, change the directory to the desktop, and then open the resulting STIX file in the Pluma editor. The Pluma editor window shows the XML content of the generated STIX file, which includes namespaces, a header with a description, and an indicator with observable data.

```
[root@parrot]~/home/analyst/openioc-to-stix
#python3 openioc-to-stix.py -i /home/analyst/openioc-to-stix/examples/ccapp
.ioc.xml -o /home/analyst/Desktop/ccap.stix.xml
[root@parrot]~/home/analyst/openioc-to-stix
#
[root@parrot]~/home/analyst/openioc-to-stix
#cd
[root@parrot]~
#cd /home/analyst/Desktop
[root@parrot]~/home/analyst/Desktop
#pluma ccap.stix.xml
```

```
ccapp.stix.xml (/home/analyst/Desktop) - Pluma (as superuser)
File Edit View Search Tools Documents Help
Open Save Undo Cut Copy Paste Find
ccapp.stix.xml x
1<stix:STIX Package
2  xmlns:cyboxVocabs="http://cybox.mitre.org/default_vocabularies-2"
3  xmlns:FileObj="http://cybox.mitre.org/objects#FileObject-2"
4  xmlns:indicator="http://stix.mitre.org/Indicator-2"
5  xmlns:stixCommon="http://stix.mitre.org/common-1"
6  xmlns:cyboxCommon="http://cybox.mitre.org/common-2"
7  xmlns:cybox="http://cybox.mitre.org/cybox-2"
8  xmlns:stix="http://stix.mitre.org/stix-1"
9  xmlns:openioc="http://openioc.org/openioc"
10 xmlns:xlink="http://www.w3.org/1999/xlink"
11 xmlns:ds="http://www.w3.org/2000/09/xmldsig#"
12 xmlns:xs="http://www.w3.org/2001/XMLSchema"
13 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
14   id="openioc:Package-decda8fa-984a-4e31-9b51-49752ee51ff4" version="1.2"
15 <stix:STIX_Header>
16   <stix:Description>CybOX-represented Indicators Translated from OpenIOC File</
stix:Description>
17 </stix:STIX_Header>
18 <stix:Indicators>
19   <stix:Indicator id="openioc:indicator-7beaf34b-3e90-477a-a5a1-laeedcd46ba6"
timestamp="2023-12-08T11:48:47.564454+00:00" xsi:type='indicator:IndicatorType'>
20     <indicator:Title>CybOX-represented Indicator Created from OpenIOC File</
indicator:Title>
21     <indicator:Observable id="openioc:item-d610019c-379f-4d3e-b299-f0b0e5c3313a">
22       <cybox:Observable_Composition operator="OR">
```

This concludes the demonstration of the structuring of collected data using **STIX**.

EXERCISE 5: Bulk Data Management and visualization

Data can be represented in a more sophisticated manner using infographics, maps, gauges and meters, detailed bars, and pie charts.

In this lab, we will perform management and visualization of collected bulk datasets. For performing this lab, we shall assume that by this stage, the analyst has collected all the required data from various sources. We shall use Tableau to manage and visualize the collected bulk data. Since the bulk data that is collected can vary depending on organizational requirements, we shall use the custom survey data related to cyber data breaches and demonstrate to manage and visualize the details regarding individuals affected as per the data feed.

Lab Objectives

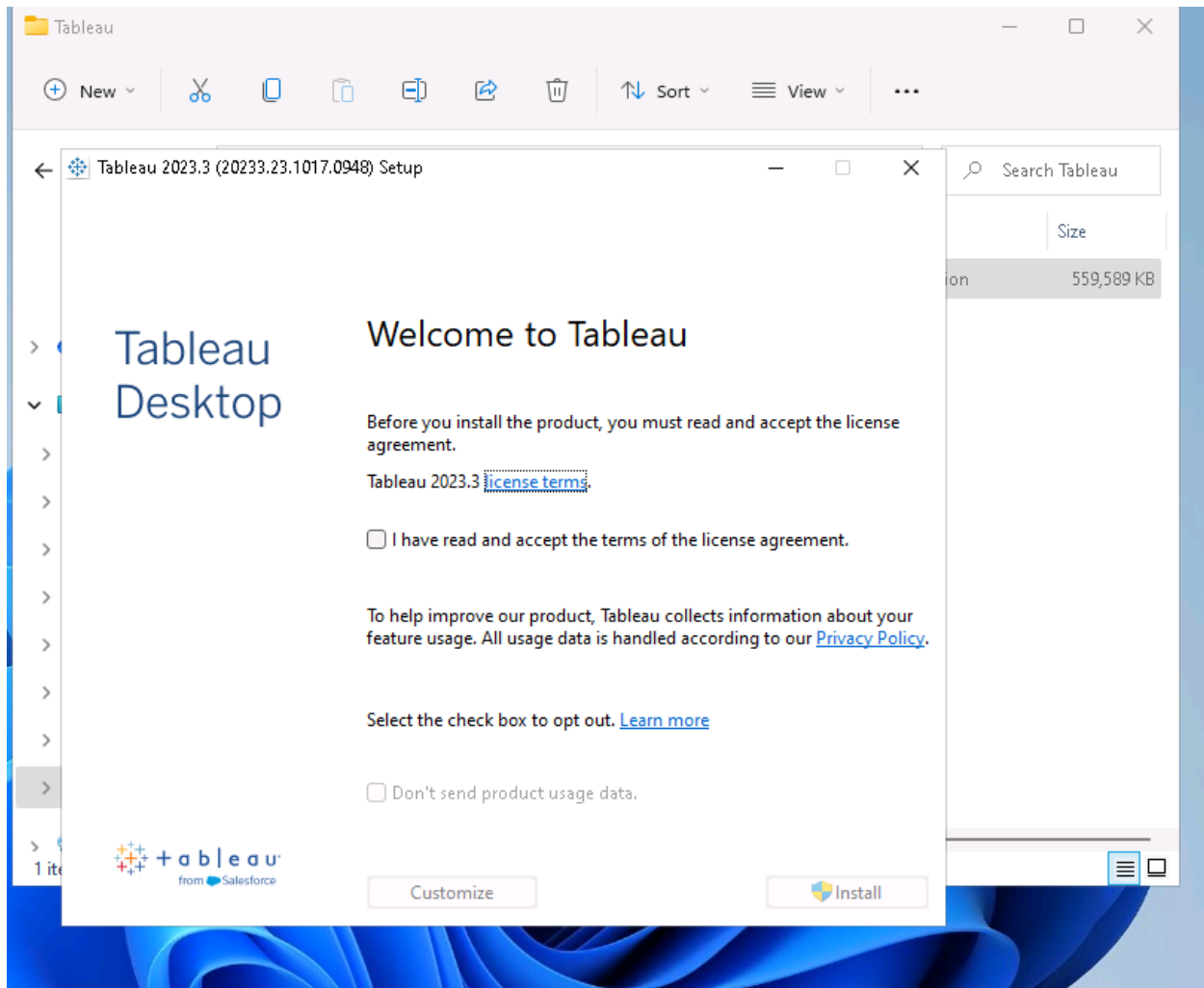
The objective of this lab is to help you understand how to manage and visualize collected bulk data using the Tableau tool.

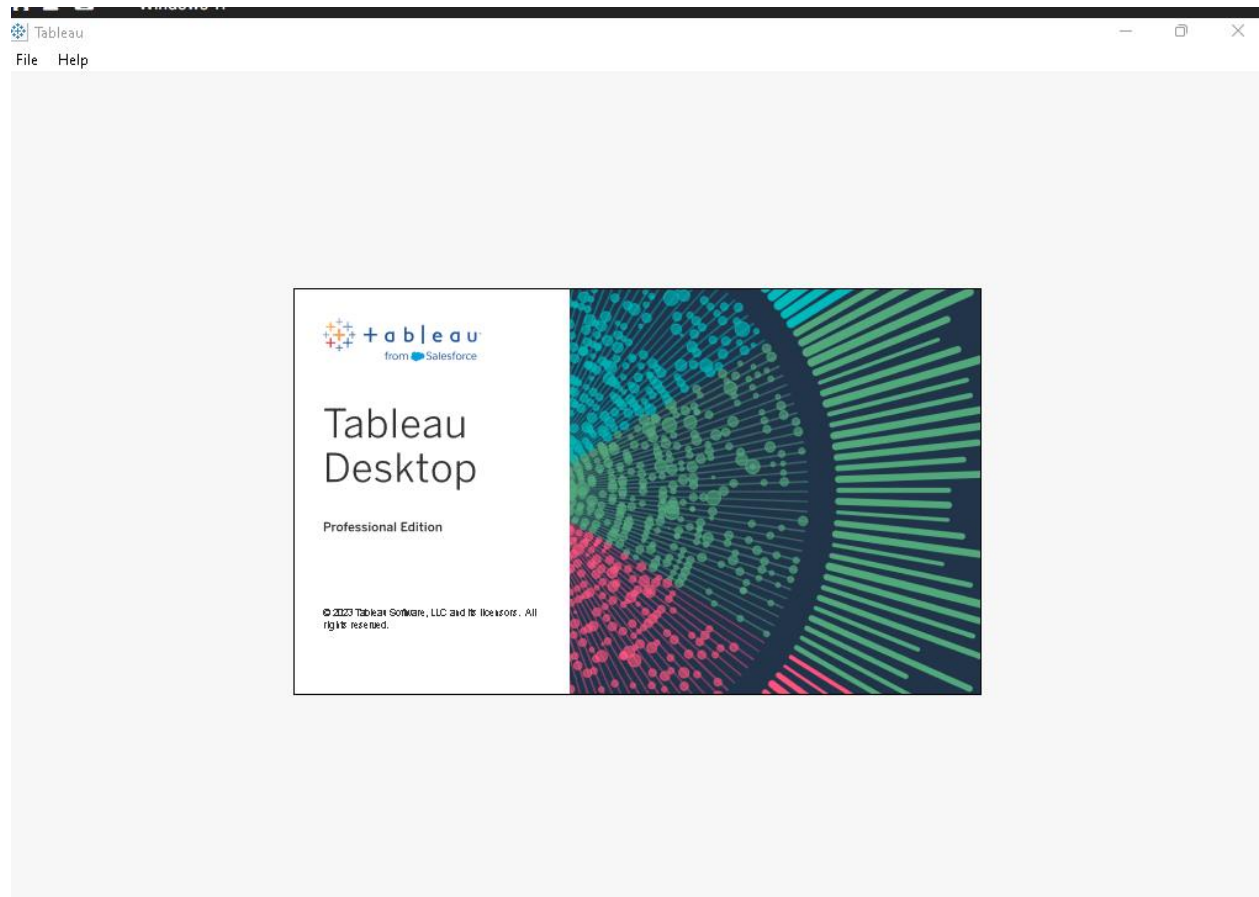
Overview of Bulk Data Management and Visualization

Data visualization is the graphical or pictorial representation of data and statistical information. It enables analysts to easily understand complex data structures, statistical details, and patterns of information that might go undetected in textual format. The visual representation is versatile enough for deeper analysis and querying of data and its manipulation, which is not possible with traditional graphs and Excel spreadsheets.

Tableau is a data visualization tool that helps people see and understand data. It is particularly efficient in handling massive and variable data sets used in big data operations with AI and machine learning. Tableau can connect to a wide variety of data sources, including databases, spreadsheets, and cloud-based applications. It can then be used to create interactive visualizations that can be shared with others.

Install Tableau





Open file using Tableau

The screenshot shows the Tableau Desktop interface. The top bar indicates 'Windows 11' and 'Tableau - Book1'. The menu bar includes 'File', 'Data', 'Server', 'Window', and 'Help'. The left sidebar has a 'Connections' section with 'HHSCyberSecurityBreaches' (Text file) and a 'Files' section with 'breaches.csv' and 'HHSCyberSecurityBreaches.csv'. The main area displays 'HHSCyberSecurityBreaches' with a 'Live' connection. A message 'Need more data? Drag tables here to relate them. [Learn more](#)' is shown. The bottom right shows a preview of the data with columns: '#', 'F1', 'Name', 'Phys...', and 'Rem...'. A 'Go to Worksheet' button is visible in the bottom left.

Windows 11

Tableau - Book1

File Data Server Window Help

Connections [Add](#)

HHSCyberSecurityBreaches
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

☒ breaches.csv

☒ HHSCyberSecurityBreaches.csv

New Union

New Table Extension

HHSCyberSecurityBreaches

Connection: ☒ Live ☐ Extract

Filters: 0 [Add](#)

HHSCyberSecurityBreach...

Need more data?
Drag tables here to relate them. [Learn more](#)

HHSCyberSecurityBreach... 10 fields 1151 rows

100 rows

Collapse

#	Abc
HHSCyberSecurityBreaches.csv	HHSCyberSecurityBreaches
F1	Name.of.Covered.Ent
1	Brooke Army Medical Center
2	Mid America Kidney Center
3	Alaska Department of Health
4	Health Services for Children
5	U.S. Department of Defense

Go to Worksheet

Add another dataset and link the two together

The screenshot shows the Tableau Desktop interface with the following components:

- Connections Panel:** Lists two text files: `HHSCyberSecurityBreaches` and `breaches`.
- Files Panel:** Shows the same two files, with a note about the Data Interpreter.
- Central Workspace:** Displays a Venn diagram indicating that `HHSCyberSecurityBreaches.csv` and `breaches.csv` share data. The connection is set to **Live**.
- Data Preview:** Shows the first few rows of the combined data. The first column is labeled `F1`.

#	HHSCyberSecurityBreaches.csv	breaches.csv
1	Brooke Army Medic	
2	Mid America Kidney	
3	Alaska Department	
4	Health Services for	

Tableau - Book1

FileDataServerWindowHelp

Home

Navigation

Refresh

Connections

Add

HHSCyberSecurityBreachesText file

breachesText file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

breaches.csv

HHSCyberSecurityBreaches.csv

New Union

New Table Extension

HHSCyberSecurityBreaches....

Connection

Live

Extract

Filters

0

Add

HHSCyberSecurityBreaches.csv is made of 2 tables.

HHSCyberSecurityBreaches.csv

breaches.csv

Join

Inner

Left

Right

Full Outer

Data Source		breaches.csv
F1	=	F1 (breaches.csv)
Add new join clause		

HHSCyberSecurityBreach...24 fields 1055 rows

100rows

Name

HHSCyberSecurityBreaches.csv

Fields

Type	Field Name	Phys...	Rem...
#	F1	HHSC...	F1

#	HHSCyberSecurityBreaches.csv	HHSCyberSecurityBreach
	F1	Name of Covered En
1		Brooke Army Medica
2		Mid America Kidney
3		Alaska Department
4		Health Services for
5		...

Go to Worksheet

Adding filter: Individuals Affected

The screenshot shows the Tableau Desktop interface with the 'HHSCyberSecurityBreaches.csv' data source loaded. The 'Filter [Individuals.Affected]' dialog box is open, allowing the user to filter data based on the number of individuals affected. The dialog is set to 'Range of values' with a range from 500 to 4,900,000. The background shows the Tableau interface with the 'HHSCyberSecurityBreaches.csv' data source and a list of breaches.

Filter [Individuals.Affected]

Range of values

500 4,900,000

500 4,900,000

☐ Include Null Values

Reset OK Cancel

Fields

Type	Field Name	Phys...	Rem...
#	F1	HHSC...	F1

Data Source

Sheet 1

HHSCyberSecurityBreaches.csv is made of 2 tables. ⓘ

Connection: ☒ Live ☐ Extract

Filters: 0 | Add

breaches.csv

HHSCyberSecurityBreaches.csv

New Union

New Table Extension

Go to Worksheet

100 rows

HHSCyberSecurityBreaches.csv	HHSCyberSecurityBreaches.csv
1	Brooke Army Medical Center
2	Mid America Kidney Center
3	Alaska Department of Health and Social Services
4	Health Services for Veterans Affairs
5	U.S. Border Patrol

Filter [Individuals.Affected]

Range of values

500 2000

500 4,900,000

☐ Include Null Values

Reset OK Cancel

Live Extract 0 | Add

Connections Add

HHSCyberSecurityBreaches
Text file

breaches
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

breaches.csv

HHSCyberSecurityBreaches.csv

New Union

New Table Extension

HHSCyberSecurityBreaches.csv is made of 2 tables. ⓘ

HHSCyberSecurityBreaches.csv breaches.csv

Join

Edit Data Source Filters

Filter	Details
Individuals.Affected	ranges from 500 to 2,000

Add... Edit... Remove

OK Cancel

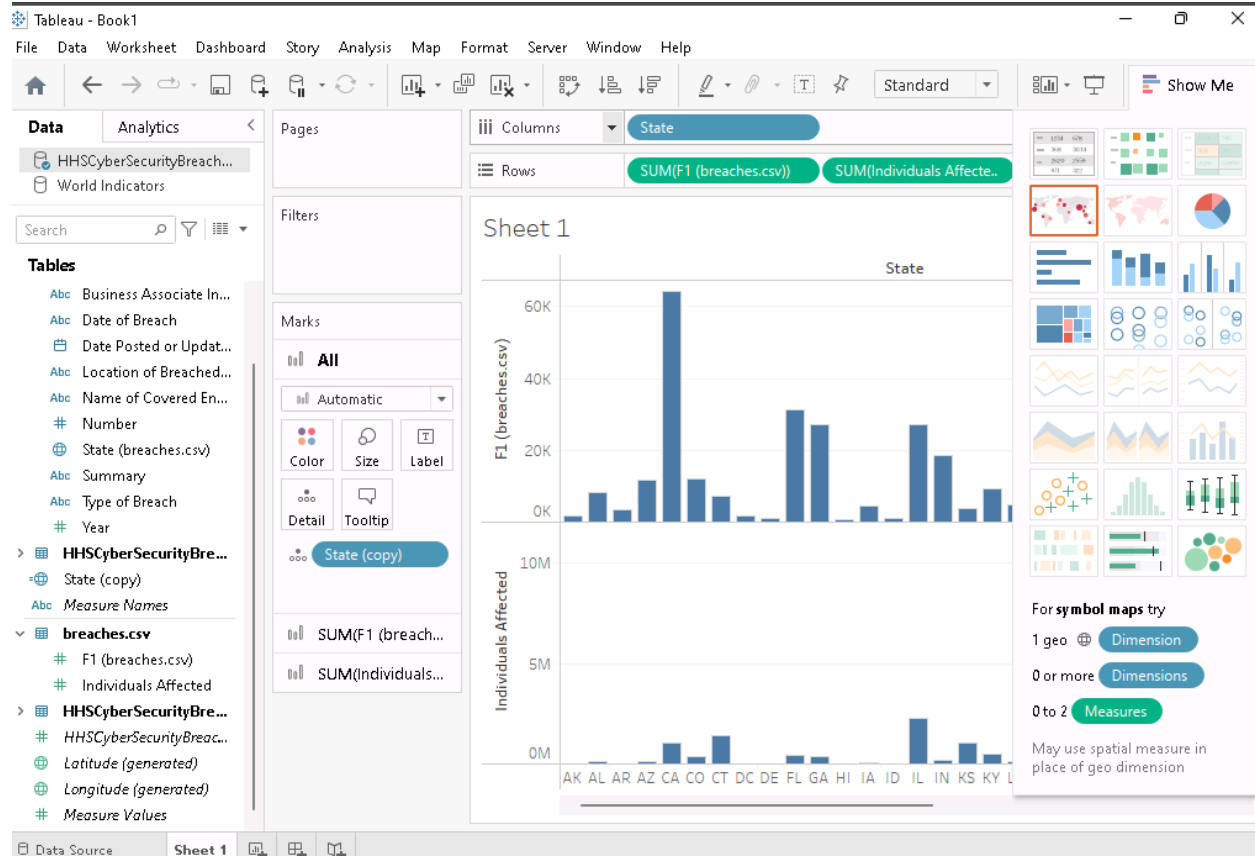
100 rows

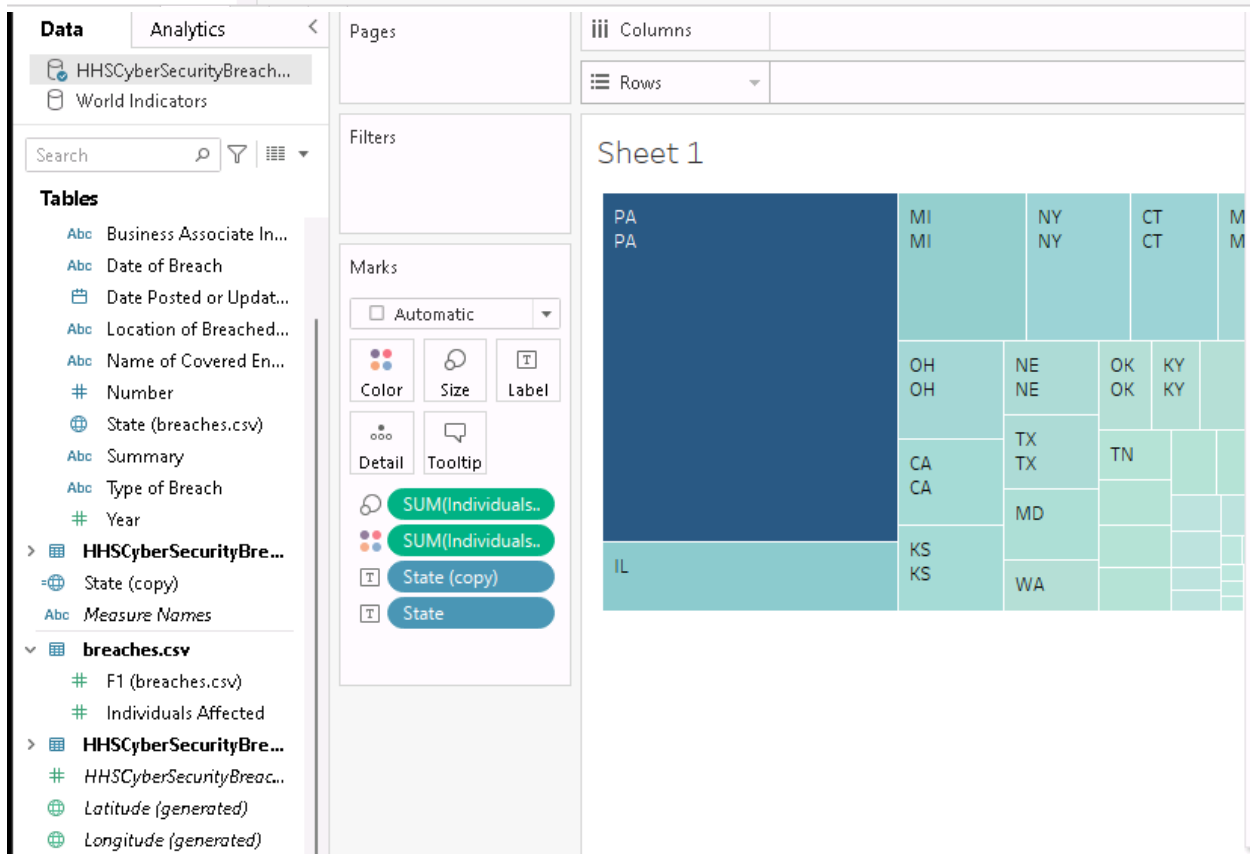
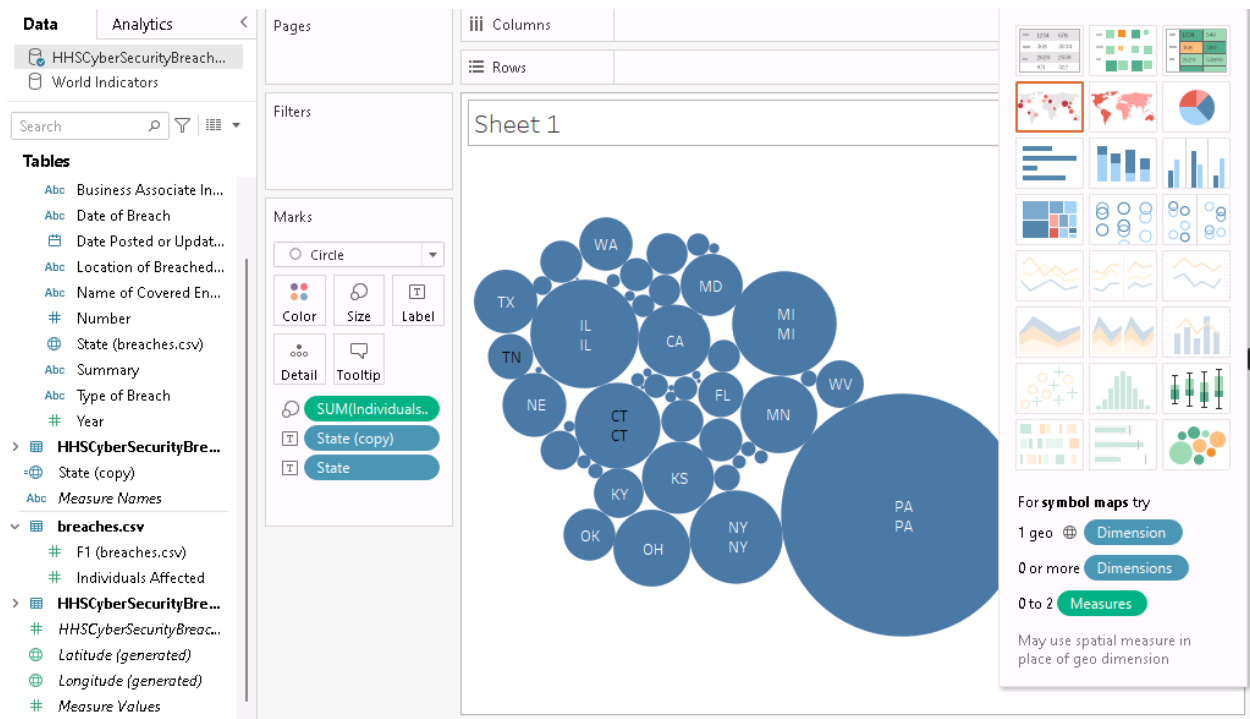
HHSCyberSecurityBreaches.csv HHSCyberSecurityBreaches.csv

Confirm Range:

You can apply many more filters and perform other analytical operations on the collected bulk datasets using the operational features available in **Tableau**. After performing the analytical operations to manage the acquired bulk dataset, we can visualize the datasets using Tableau. First, we need to **remove all** the applied filters for the imported datasets. To do this, click on **Edit** option under **Filters** present at the top-right corner of the window. **Edit Data Source Filters** window appears, select the applied filter in the **Edit Data Source Filters** window and click **Remove** and then **OK**. The applied filter will be removed.

Showing different state of the dataset





This concludes the demonstration of bulk data management and visualization using Tableau

